



**UNIVERSITAT POLITÈCNICA
DE CATALUNYA
BARCELONATECH**



**telecos
BCN**



ACCESS TO UNDERGRADUATE PROGRAMMES IN CATALAN UNIVERSITIES: PATTERN ANALYSIS, GAP IDENTIFICATION AND PREDICTIVE SIMULATION

NOA MEDIAVILLA SOUTHWOOD

Thesis supervisor

GUILLEM PERARNAU LLOBET (Department of Mathematics)

Degree

Bachelor's Degree in Data Science and Engineering

Bachelor's thesis

Facultat d'Informàtica de Barcelona (FIB)

Escola Tècnica Superior d'Enginyeria de Telecomunicació de Barcelona (ETSETB)

Facultat de Matemàtiques i Estadística (FME)

Universitat Politècnica de Catalunya (UPC) - BarcelonaTech

27/06/2025

Acknowledgments

To Guillem, thank you for your support, trust and kindness throughout the thesis. Your guidance has been invaluable to me.

To Blanca, Maria, Andrea, Núria, Laia and Rebeca, thank you for your endless patience and for being by my side every step of the way for the past four years. I don't know if this journey would have been possible without your friendship and encouragement.

To Jana and Nina, my biggest supporters since day one, thank you for always believing in me and cheering me on, no matter the circumstances.

Finally, to my mother, who has taught me the true value of education and has always put my future before her own dreams. Thank you for your unconditional love and sacrifices. I would not be where I am today without you.

Abstract

This thesis investigates the landscape of university admissions in Catalonia, focusing on the interplay between educational pathways, socioeconomic backgrounds, and the structure of student preferences. Leveraging a rich, multi-source dataset, including application records, admissions outcomes, and enriched socioeconomic indicators, this work applies network science techniques, clustering, and recommender system methodologies to uncover hidden patterns and dynamics in higher education access.

The study begins by contextualizing the current educational system and the various routes students take to enter university, drawing on recent research and statistical data. It then formulates a central hypothesis: that socioeconomic factors and institutional structures significantly shape both the opportunities available to students and the diversity of student profiles.

A comprehensive technical framework is developed, encompassing temporal network analysis, graph theory, and clustering techniques, in order to build a knowledge graph that models the relationships between degrees based on application co-occurrences. This knowledge graph is analyzed over multiple years to reveal evolving clusters, persistent niches, and the stability or transformation of academic fields.

The thesis further explores the socioeconomic dimension by mapping student flows from secondary schools to university degrees, analyzing geographic and income-based patterns and quantifying the diversity of student backgrounds within each degree.

Finally, the project culminates in the development of a simple recommender system designed to empower prospective students. By providing personalized suggestions and interactive exploration tools, the system aims to broaden students' horizons and promote more equitable access to higher education.

Through this multifaceted approach, the thesis not only illuminates the structural and social variables at play in university admissions but also proposes practical tools to foster greater diversity and opportunity within the academic landscape.

Resum

Aquesta tesi investiga el panorama de les admissions universitàries a Catalunya, centrant-se en la interacció entre els itineraris educatius, els orígens socioeconòmics i l'estructura de les preferències estudiantils. Aprofitant un ric conjunt de dades de múltiples fonts, que inclou registres de sol·licituds, resultats d'admissió i indicadors socioeconòmics enriquits, aquest treball aplica tècniques de ciència de xarxes, *clustering* i metodologies de sistemes de recomanació per descobrir patrons i dinàmiques ocultes en l'accés a l'educació superior.

L'estudi comença contextualitzant el sistema educatiu actual i les diverses vies que els estudiants poden seguir per accedir a la universitat, basant-se en recerques recents i dades estadístiques. A continuació, formula una hipòtesi central: que els factors socioeconòmics i les estructures institucionals influeixen significativament tant en les oportunitats disponibles per als estudiants com en la varietat de perfils d'estudiants.

Es desenvolupa un marc tècnic integral que abasta l'anàlisi de xarxes temporals, la teoria de grafs i tècniques de *clustering*, per tal de construir un graf de coneixement que modela les relacions entre titulacions a partir de la co-ocurrència en les sol·licituds. Aquest graf de coneixement s'analitza al llarg de diversos anys per revelar l'evolució dels clústers, nínxols persistents i l'estabilitat o transformació dels camps acadèmics.

La tesi explora també la dimensió socioeconòmica *mapejant* els fluxos d'estudiants des dels centres de secundària fins als graus universitaris, analitzant patrons geogràfics i basats en la renda, i quantificant la diversitat d'orígens estudiantils dins de cada titulació.

Finalment, el projecte culmina amb el desenvolupament d'un senzill sistema de recomanació dissenyat per empoderar els futurs estudiants. Proporcionant suggeriments personalitzats i eines interactives d'exploració, el sistema pretén ampliar els horitzons dels estudiants i promoure un accés més equitatiu a l'educació superior.

Amb aquest enfocament multifacètic, la tesi no només il·lumina les variables estructurals i socials que intervenen en les admissions universitàries, sinó que també proposa eines pràctiques per fomentar una major diversitat i oportunitats dins l'àmbit acadèmic.

Resumen

Esta tesis investiga el panorama de las admisiones universitarias en Cataluña, centrándose en la interacción entre las trayectorias educativas, los orígenes socioeconómicos y la estructura de las preferencias estudiantiles. Aprovechando un rico conjunto de datos de múltiples fuentes, que incluye registros de solicitudes, resultados de admisión e indicadores socioeconómicos enriquecidos, este trabajo aplica técnicas de ciencia de redes, agrupamiento y metodologías de sistemas de recomendación para descubrir patrones y dinámicas ocultas en el acceso a la educación superior.

El estudio comienza contextualizando el sistema educativo actual y las diversas vías que los estudiantes pueden tomar para acceder a la universidad, basándose en investigaciones recientes y datos estadísticos. A continuación, formula una hipótesis central: que los factores socioeconómicos y las estructuras institucionales influyen significativamente tanto en las oportunidades disponibles para los estudiantes como en la diversidad de los perfiles de estudiantes.

Se desarrolla un marco técnico integral que abarca el análisis de redes temporales, la teoría de grafos y técnicas de agrupamiento, con el fin de construir un grafo de conocimiento que modela las relaciones entre titulaciones a partir de la co-ocurrencia en las solicitudes. Este grafo de conocimiento se analiza a lo largo de varios años para revelar la evolución de los clústeres, nichos persistentes y la estabilidad o transformación de los campos académicos.

La tesis explora además la dimensión socioeconómica *mapeando* los flujos de estudiantes desde los centros de secundaria hasta los grados universitarios, analizando patrones geográficos y basados en la renta, y cuantificando la diversidad de orígenes estudiantiles en cada titulación.

Finalmente, el proyecto culmina con el desarrollo de un sencillo sistema de recomendación diseñado para empoderar a los futuros estudiantes. Al proporcionar sugerencias personalizadas y herramientas interactivas de exploración, el sistema pretende ampliar los horizontes de los estudiantes y promover un acceso más equitativo a la educación superior.

A través de este enfoque multifacético, la tesis no solo ilumina las variables estructurales y sociales que intervienen en las admisiones universitarias, sino que también propone herramientas prácticas para fomentar una mayor diversidad y oportunidades en el ámbito académico.

Contents

1	Introduction	6
2	Technical Background	9
2.1	Graph Theory and Network Analysis	9
2.2	Community Detection and Clustering Techniques	10
2.3	Recommender Systems and ML Models	12
2.4	Evaluation and Distance Metrics	14
3	Preprocessing	16
4	Data Analysis: Degree Knowledge Graph	18
4.1	Initial Approach: Degree Programs as Probability Distributions Based on Application Preferences	18
4.2	Transition to NLP-Inspired Analysis: Frequencies and Co-Occurrences	20
4.3	Initial Knowledge Graph Definition: Parameters and Rationale	21
4.4	Refining the Model: Parameter Adjustments and Enhanced Visualization	24
4.5	Temporal Evolution of the Knowledge Graph	27
4.5.1	Cluster Strengthening and Dispersion: Centrality Measures and Annual Graph Statistics	27
4.5.2	Heatmap Representations: Cluster Stability Over Time	30
4.5.3	Top Co-occurrences: Identifying Persistent Niches	32
4.6	Clustering the Knowledge Graph	35
4.6.1	Community Detection and Clustering Methods	35
4.6.2	Benchmarking Against UPC's Classification	39
5	Data Analysis: Flows and Socioeconomic Patterns	43
5.1	Variable Analysis Relative to Assigned Center	43
5.1.1	Principal Component Analysis	43
5.1.2	Correlation Analysis	44
5.2	Mapping Socioeconomic Flows	46
5.2.1	Flows from Secondary Schools to University Degrees: Quantifying Transitions	46
5.2.2	Geographic Flows: Tracking Student Movement by Postal Code Over Time	48
5.2.3	Linking Family Income Index to Degrees: Identifying High-Income Degrees, Annual Rankings and Social Niches	51
5.3	Interpreting Patterns: Dynamic, Static, and Hermetic Degree Profiles	53
6	Recommender Systems	55
6.1	First Versions	55
6.2	Simple Machine Learning Models	57
7	Conclusions and Future Work	60
7.1	Future Work: Complex Model Proposals	60
7.1.1	Encoder–Decoder Architectures	60
7.1.2	Graph Neural Networks	60
7.2	Conclusions	61

1 Introduction

The educational system in Catalonia is designed to provide a structured and flexible pathway for students, ensuring the younger part of the population can have access to lifelong learning and a variety of professional and academic opportunities.

Governed by both national and regional legislation, such as the Organic Law of Education (LOE) of 2006 [1], its subsequent modification (LOMLOE) of 2020 [2], and the Catalan Education Law (LEC) of 2009 [3], the system is organized into distinct stages: from early childhood education through to university and specialized professional training. Each stage is designed to provide smooth transitions, allowing students to re-enter the system at different points through adult education centers, vocational training for employment, or specific entrance exams.

In Catalonia, educational attainment levels reflect a diverse and dynamic population. The system begins with voluntary early childhood education (ages 0-6), followed by an obligatory primary education (ages 6-12), which can be provided for free by the state, and then the compulsory secondary education (ESO) from ages 12 to 16. Upon completion of ESO, students can choose between the academic route of baccalaureate *Batxillerat* or the more professionally oriented vocational training (*Formació Professional*, FP), both of which open doors to higher education or direct entry into the labor market. The baccalaureate, typically undertaken between ages 16 and 18, is a crucial stage for those aiming to access university studies, while vocational training offers multiple levels and specializations, catering to a wide range of professional aspirations. [8, 10]

A notable feature of the Catalan educational landscape is the diversity of school types: public, private, and semi-private centers. Public schools are state-funded and secular, with admissions regulated by the Department of Education. Private schools are funded by families and have greater autonomy in admissions and management. Semi-private schools, while privately managed, receive public subsidies and must adhere to certain public regulations. The allocation of places in *Batxillerat* programs, for example, is governed by a transparent, points-based system that prioritizes students based on established legal criteria, such as affiliation between schools and specific priority or complementary factors.

To fully understand the educational landscape in Catalonia, it is essential to consider the influence of territorial socioeconomic disparities. The Socioeconomic Territorial Index (IST) provides a comprehensive measure of these differences, synthesizing multiple dimensions, such as employment, education, immigration, and income, into a single, comparative value. With a reference value of 100 for Catalonia as a whole, the IST allows for a nuanced mapping of inequality across the region's 947 municipalities, as well as within smaller census areas and neighborhoods. [12]

The 2021, IST data revealed an uneven distribution of socioeconomic resources. Municipalities such as Matadepera, Castellolí, Sant Just Desvern, Tiana, and Sant Cugat del Vallès stand out with the highest IST scores, exceeding the Catalan average by more than 20 percentage points. These areas, concentrated in the Central Counties, the Metropolitan Area, and the Girona region, consistently outperform in all six IST indicators: higher employment rates, lower prevalence of low-skilled workers, higher educational attainment, lower rates of young people without post-compulsory studies, lower levels of immigration, and higher household incomes. In contrast, municipalities in Ponent and other parts of Girona, such as Sant Climent Sescebes and Querol, register IST values more than 20 points below the average, reflecting persistent disadvantage across all measured dimensions. [16]

A particularly telling indicator is the proportion of young people without post-compulsory education, which exhibits the greatest disparity between municipalities. In some localities, such as Fontanals de Cerdanya and Matadepera, this figure is as low as 1.3% and 5.3%, respectively, while in others, like Barbens and La Jonquera, it soars above 57%. [15] This gap is not merely statistical, it signals the existence of entrenched barriers to educational progression that are closely tied to local socioeconomic conditions. The IST not only quantifies inequality but also highlights the mechanisms by which it is reproduced: access to quality education, stable employment, and information about opportunities is systematically more limited in disadvantaged areas.

Catalonia's population of over 6.6 million people aged 15 and above is increasingly well-educated, with more than a third (34.7%) having completed higher education. However, 14% remain at the level of primary education or below, and the majority are concentrated in the secondary education stages. This distribution reflects both generational change and the enduring centrality of secondary education as the formative core of the population. [15]

The evolution of university enrollment and graduation in Catalonia further illustrates the interplay between demographic, economic, and gender factors. Over the past two decades, university enrollment has shown remarkable resilience and steady growth, particularly in comparison to the more volatile trends observed in Spain as a whole. [13] Periods of economic crisis have tended to boost demand for higher education, as young people seek to delay entry into a difficult labor market, while subsequent recoveries have seen a partial shift towards vocational training or direct employment.

A striking feature of this evolution is the sustained and widening feminization of university enrollment. Since 2001, women have consistently outnumbered men among university students, with the gap reaching nearly 30,000 by 2022. [13] This trend is not only quantitative. Women also exhibit higher success rates and lower dropout rates, although horizontal segregation persists, with women still underrepresented in engineering and technology fields. The recent surge in enrollment, driven by demographic factors (such as the entry of the 2003-2005 baby boom group) and the appeal of double degrees and international programs, has placed new demands on university resources and highlighted the need for flexible, student-centered approaches.

Graduation rates have mirrored enrollment trends, with periods of rapid increase followed by stabilization. The transition to the Bologna system, economic cycles, and the impact of the COVID-19 pandemic have all contributed to fluctuations in the number of graduates. In particular, the volatility in graduation numbers can strain regulated professions and underscores the importance of real-time labor market monitoring and the promotion of paid internships to better align educational output with economic needs. [14]

Given the circumstances, it is essential to address a critical and often contentious dimension of educational equity: the role of grade inflation and the differential opportunities it creates between students from public and private high schools. Recent analyses of university entrance data in Spain, and particularly in Catalonia, have brought to light a persistent and significant disparity in the grades awarded by different types of secondary schools. [20] This phenomenon is not merely a technicality in the calculation of admission scores, it is a structural factor that shapes the very landscape of access to higher education, especially to the most competitive and in-demand programs.

The data revealed a clear pattern: students from private, and especially non-subsidized private, high schools are disproportionately represented among those with the highest high school grades. On the other hand, while the average high school grade for students deemed eligible for university admission stands at 7.77, the average score on the general phase of the university entrance exam (PAU) drops to 6.81, with nearly 10% of candidates failing to meet the threshold. This gap is not uniform across school types. In public schools, a majority of students (almost 60%) receive high school grades below 8, whereas in private schools (both subsidized and non-subsidized) the proportion of students with grades between 9 and 10 is markedly higher (27.4% in non-subsidized private schools, 23.9% in subsidized, compared to just 17.9% in public schools). [20]

This inflation of high school grades in private institutions has a direct and substantial impact on university admissions. The result is a double inequity: first, students from private schools are more likely to access the most sought-after university programs, not necessarily because of superior academic ability, but because of the grading practices of their schools; second, these disparities are compounded by regional differences in grading and exam correction criteria, which, under the current system of a unified admissions district, allow students from certain regions to compete for places in universities across Spain, sometimes to the disadvantage of local applicants.

Beyond the numbers, this situation raises profound questions about the fairness and transparency of the university admissions process. It challenges the assumption that merit, as measured by grades, is an objective and equalizing standard. Instead, it exposes how institutional practices and regional policies can reinforce existing social and economic divides, privileging those with access to certain types of schools

or regions over others.

Yet, the aim of this thesis arises under the hypothesis that the issue of equity in university access extends beyond the mechanics of grade calculation. We believe that there is also a subtler, but equally powerful, layer of inequality rooted in the dissemination of information and the socialization of academic aspirations. Not all students are equally aware of the full spectrum of academic opportunities available to them. Certain degree programs, and especially those in emerging fields like Data Science, Artificial Intelligence, or cutting-edge technological disciplines, often remain the preserve of those who, through family, school, or social networks, are more closely connected to the worlds where such knowledge circulates.

This phenomenon creates what might be termed "hidden curricula of opportunity": some career paths are widely known and actively promoted within certain social environments, while others are transmitted only through informal, word-of-mouth channels, remaining largely invisible to students from less advantaged backgrounds or from schools with fewer resources and weaker ties to the professional or technological sectors. As a result, the reproduction of professional elites and the concentration of talent in specific fields are not only the product of academic performance, but also of the uneven distribution of information and guidance.

Therefore, this thesis seeks to explore the deeper networked structures that govern who learns about, aspires to, and ultimately accesses the most dynamic and future-oriented academic programs. By analyzing admissions to public university centers and programmes in Catalonia and by employing tools from graph theory and network analysis, this research will explore how patterns of application and admission are shaped not only by grades and formal criteria, but also by the flows of information, the density of social connections, and the institutional pathways that channel students toward or away from certain opportunities.

All results, additional material, and code built to obtain the presented results can be found at:
<https://github.com/Noameso/equitable-ds-access-cat>

2 Technical Background

2.1 Graph Theory and Network Analysis

Graph Theory Graph Theory is the branch of discrete mathematics that studies graphs: ordered pairs $G = (V, E)$ in which V is a finite set of vertices (or nodes) and $E \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is a set of edges (or links) joining pairs of vertices.¹ A graph may be undirected (edges have no orientation) or directed (arcs have a source and a target), simple (no multiple edges or loops) or a multigraph, and weighted or unweighted. Fundamental constructs such as degree, paths, cycles, connectivity, and sub-graphs allow one to formalise questions about reachability, flow and optimisation of these structures. Because these abstractions appear in chemistry (molecular bonds), computer science (data structures, communication networks), biology (metabolic networks), and many other fields, Graph Theory provides a unifying mathematical language for a wide array of scientific problems. [7]

Network Analysis Network Analysis (or network science) applies graph-theoretic ideas to empirical systems in order to measure, model, and predict the behaviour of real-world networks. Given an observed network represented by its adjacency matrix $A = (a_{ij})$, the analysis seeks to (i) characterise structure through metrics such as degree distribution $P(k)$, average path length $\langle \ell \rangle$, clustering coefficient C , and node centralities; (ii) detect meso-scale patterns like communities or core-periphery structures; and (iii) simulate dynamical processes (e.g. information diffusion, epidemic spread) that unfold over the links of the graph. In practice, network analysis turns raw relational data (from social interactions and transportation grids to gene-regulation maps) into quantitative insights about robustness, influence, optimisation, and control, thereby translating the abstract machinery of Graph Theory into actionable knowledge for disciplines across the natural and social sciences. [24, 38]

Degree centrality In a simple, undirected graph $G = (V, E)$ with $n = |V|$ nodes, the degree centrality of a node $v \in V$ is the number of incident edges,

$$C_D(v) = \deg(v), \quad C_D^{\text{norm}}(v) = \frac{\deg(v)}{n-1}.$$

Degree centrality captures the local exposure of v : a high-degree node can directly exchange information or resources with many neighbours in a single step, making it an effective immediate hub. [29]

Closeness centrality For a connected graph, the closeness centrality of v is the inverse of its total distance to all other nodes,

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} d(v, u)},$$

where $d(v, u)$ is the length of the shortest path between v and u . A node with high closeness is, on average, only a few hops away from every other node, so it can disseminate (or receive) information quickly throughout the entire network. [26]

Betweenness centrality Given the set of all shortest paths, the betweenness centrality of v is

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths from s to t and $\sigma_{st}(v)$ is the count of those paths that pass through v . Nodes with high betweenness lie on many geodesic routes and thus function as strategic brokers or bridges that can control, filter, or facilitate flows across otherwise distant parts of the network. [28]

Eigenvector centrality Let A be the adjacency matrix of G . Eigenvector centrality assigns each node v a score proportional to the scores of its neighbours:

$$x_v = \frac{1}{\lambda} \sum_t a_{vt} x_t \iff A\mathbf{x} = \lambda\mathbf{x}.$$

¹For directed graphs the edge set is a subset of ordered pairs, $E \subseteq V \times V$. Weighted graphs add a map $w : E \rightarrow R$ that assigns a real-valued cost, length, or capacity to each edge.

The vector $\mathbf{x}$ is the dominant eigenvector (Perron-Frobenius) of A . This recursive definition embodies the idea that a node is influential when it is connected to other influential nodes, so importance cascades through the network. [23]

PageRank PageRank refines eigenvector centrality for directed graphs with a "random-surfer" model. With damping factor $\alpha \in (0, 1)$ and out-degree $L(j) = \sum_k a_{jk}$, the PageRank of node i satisfies

$$p_i = \alpha \sum_j a_{ji} \frac{p_j}{L(j)} + \frac{1 - \alpha}{n},$$

or in matrix form $\mathbf{p} = \alpha \mathbf{T}^\top \mathbf{p} + \frac{1 - \alpha}{n} \mathbf{1}$, where $\mathbf{T}$ is the row-stochastic transition matrix. The steady-state probability p_i represents the fraction of time a random surfer spends on i . Therefore, nodes with high PageRank enjoy high global visibility that balances popularity with the likelihood of random "teleportation" to any page. [4]

Modularity For a given partition of an undirected (possibly weighted) graph $G = (V, E)$ into communities $\{c_1, c_2, \dots, c_k\}$, modularity Q compares the density of intra-community edges with the density expected under the configuration-model null hypothesis:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j),$$

where A_{ij} is the (weighted) adjacency matrix, $k_i = \sum_j A_{ij}$ is the degree (strength) of node i , $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is the total edge weight, and δ is the Kronecker delta. Q is positive when communities contain more internal edges than expected at random, negative when they contain fewer, and typically lies in the interval $[-1, 1]$. Hence Q quantifies how strongly the network decomposes into self-contained modules: the closer Q is to its theoretical maximum, the sharper the community structure. [36]

Normalised modularity Because the maximum attainable modularity $Q_{\max}$ depends on the network's size and degree sequence, many studies rescale Q to allow comparisons across graphs:

$$Q_{\text{norm}} = \frac{Q}{Q_{\max}}, \quad \text{with } Q_{\max} = 1 - \sum_c a_c^2, \quad a_c = \frac{1}{2m} \sum_{i \in c} k_i.$$

$Q_{\text{norm}} \in [0, 1]$ measures the fraction of the theoretically possible modularity "realised" by the chosen partition, so networks of different scales can be ranked on a common footing. Some authors further subtract the average modularity Q_{rand} of degree-preserving randomisations, $Q_{\text{norm}} = \frac{Q - Q_{\text{rand}}}{Q_{\max} - Q_{\text{rand}}}$, to discount spurious structure present even in random graphs. In either case, a value near 0 indicates community structure no stronger than random expectation, whereas $Q_{\text{norm}} \approx 1$ signals communities that are almost as sharp as theoretically possible for that degree sequence. [19]

2.2 Community Detection and Clustering Techniques

Large-scale networks, whether social, biological, or technological, often exhibit a mesoscopic structure in which vertices partition into relatively dense subgraphs called communities or clusters.

Uncovering these modules reveals functional organisation (protein complexes, interest groups, topical Web pages), improves downstream tasks such as link prediction, and reduces computational complexity by enabling coarse-grained reasoning. Conceptually, community detection extends classical clustering to graphs, where similarity is encoded by edges rather than metric distances; yet the two domains share common goals: maximise intra-group affinity and minimise inter-group connectivity.

Over the past two decades researchers have proposed a rich palette of algorithms ranging from heuristic greedy searches for modularity optima, through spectral embeddings that relax discrete partitions to continuous eigenvector spaces, to probabilistic generative models that treat clusters as latent variables. Because no single objective captures all notions of a "good" community, practitioners must weigh trade-offs among scalability, robustness to resolution limits, ability to handle weighted or directed edges, and suitability for overlapping or hierarchical structures.

Label-Propagation Communities Label-propagation detects communities by letting node labels "diffuse" over the network until a consensus emerges. At iteration $t+1$ each node adopts the most frequent label among its neighbours,

$$\ell_i^{(t+1)} = \text{mode}_{j \in \mathcal{N}(i)} \{ \ell_j^{(t)} \},$$

breaking ties uniformly at random. Dense subgraphs stabilise quickly because their internal agreement overpowers conflicting external labels, so the fixed point partitions the graph into communities with high internal density and sparse external ties. [27]

Greedy Modularity Optimisation The Clauset-Newman-Moore algorithm begins with each vertex in its own community and repeatedly merges the pair (C_i, C_j) that produces the largest modularity gain

$$\Delta Q_{ij} = \frac{e_{ij} - 2a_i a_j}{2m},$$

where e_{ij} is the fraction of edges joining the two communities, $a_i = \sum_k e_{ik}$ is the degree fraction of C_i , and m is the total edge count.

The greedy sequence stops when all remaining merges would decrease Q , yielding a hierarchy of partitions from fine to coarse that approximates the global maximum of modularity in near-linear time for sparse graphs.

Louvain Communities The Louvain method iteratively maximises modularity in two phases.

First, each node i is tentatively moved to the neighbouring community C that yields the highest positive

$$\Delta Q_{i \rightarrow C} = \frac{k_{i,\text{in}}}{m} - \frac{k_i \Sigma_C}{(2m)^2},$$

with $k_{i,\text{in}}$ the weight of edges from i into C and Σ_C the sum of degrees in C .

Second, the newly found communities are collapsed into "super-nodes" and the procedure repeats on the induced graph.

The multilevel pass ends when no individual move improves modularity, producing a fast, scalable partition with high Q . [35]

Spectral Clustering (Laplacian + k -means) Spectral Clustering follows a two-stage algorithm:

- Given a similarity (adjacency) matrix A , we construct the (unnormalised) graph Laplacian $L = D - A$ with degree matrix $D = \text{diag}(d_i)$.
- We then compute the matrix $U \in \mathbb{R}^{n \times k}$ whose columns are the eigenvectors associated with the k smallest non-zero eigenvalues of L .
- Finally, we normalise the rows and apply k -means to the resulting points $\{u_i^\top\}$.

Eigenvectors provide a low-dimensional embedding that separates sparsely connected subgraphs, while the final k -means step assigns points into k clusters that respect the graph's spectral geometry. [34, 41]

Graph Laplacian Matrix For an undirected simple graph $G = (V, E)$ with adjacency A and degree matrix D , the Laplacian is

$$L = D - A, \quad L_{ij} = \begin{cases} d_i & i = j, \\ -1 & i \neq j \text{ and } (i, j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

L is positive-semidefinite, has a zero eigenvalue corresponding to the all-ones vector, and its spectral properties govern diffusion, random walks, and cuts on the graph, hence its central role in spectral clustering and community detection.

The graph Laplacian functions as a discrete second derivative on a network: when applied to a node-valued signal, it outputs the net imbalance between a node's value and the mean of its neighbours, spotlighting where "flow" would occur. Because its eigenvectors with the smallest non-zero eigenvalues vary most smoothly across the graph, L simultaneously governs diffusion processes and reveals the sparse cuts (communities) that make such smooth variation hardest to maintain.

***k*-means Clustering** *k*-means partitions a set of vectors $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ into *k* clusters by minimising the within-cluster sum of squares

$$\min_{\{C_r\}, \{\mu_r\}} \sum_{r=1}^k \sum_{x_i \in C_r} \|x_i - \mu_r\|_2^2,$$

where $\mu_r = \frac{1}{|C_r|} \sum_{x_i \in C_r} x_i$ is the centroid of cluster C_r .

The Lloyd iteration alternates between assigning each point to its nearest centroid and recomputing centroids until assignments stabilise, converging to a local optimum of the objective. [34]

Agglomerative Clustering Agglomerative hierarchical clustering starts with each observation as a singleton cluster and repeatedly merges the two clusters A, B that minimise a linkage distance $d_{\text{link}}(A, B)$ (e.g. single-link, complete-link, average-link or Ward’s criterion). The process continues until all points belong to one cluster, generating a nested series of partitions that can be visualised as a dendrogram.

Hierarchical Clustering More generally, hierarchical clustering builds a tree T whose leaves are individual data points and whose internal nodes represent unions of their child clusters. Two inverse strategies exist: bottom-up agglomerative (dominant in practice) and top-down divisive. Cutting T at height h yields a flat clustering, while the entire tree retains multiscale information useful for exploratory analysis. [9, 31]

Dendrogram A dendrogram is a planar, tree-like diagram that plots the sequence of merges (or splits) in hierarchical clustering. The horizontal axis lists objects/clusters, and the vertical axis encodes the dissimilarity or linkage distance at which clusters join. The height of each internal node therefore quantifies cluster cohesion, and horizontal ”cuts” translate this visual summary into partitions at any desired granularity. [30]

2.3 Recommender Systems and ML Models

Recommender systems are decision engines that learn personalised rankings of items, such as films, news, or products, by mining large-scale logs of user behaviour and rich side information. Their core objective is to predict the next interaction that maximises user satisfaction or business utility while coping with sparsity, cold-start users, non-stationary preferences and real-time latency constraints. Traditionally, collaborative-filtering methods such as matrix factorisation have dominated. However, the modern landscape now blends heterogeneous data sources (text, images, social ties, knowledge graphs) and therefore demands a toolbox of machine-learning models that can capture non-linear patterns, long-range sequential signals, and complex relational structures. [6, 21]

We examine five representative families of models that have proved effective in contemporary recommender pipelines. We start with Random Forests, a bagging-based ensemble that delivers strong baseline accuracy and interpretable feature importance. Next, Multinomial Logistic Regression (softmax regression) provides a light-weight, convex alternative for scenarios requiring rapid online updates. We then introduce Singular Value Decomposition (SVD), the latent-factor engine behind the Netflix Prize and countless production systems. Moving to deep learning for future work, we outline Encoder–Decoder architectures that treat recommendation as a sequence-prediction task, and finally, Graph Neural Networks (GNNs) that leverage the high-order collaborative signals inherent in user-item interaction graphs.

Random Forest Classifier A decision tree is a flow-chart-shaped predictive model in which every internal node poses a yes/no (or multi-way) question about one input feature, the outgoing branches correspond to the possible answers, and each leaf node stores the final prediction, typically a class label for classification or a numeric value for regression.

A Random Forest is an ensemble of decision trees: each tree is built from a random slice of the training rows and a random subset of the columns (features), therefore it bootstraps samples of the data and splits them using randomly selected feature subsets.

When a new example x arrives, every tree gives a class vote and the forest predicts the class that wins the majority,

$$\hat{y} = \arg \max_c \sum_{b=1}^B \mathbf{1}[T_b(x) = c].$$

By following the path defined by an example's feature values from the root to a leaf, the model maps $R^d \rightarrow \{1, \dots, C\}$ (classification) or $R^d \rightarrow R$ (regression) using a sequence of simple, interpretable tests. Because the trees have seen different data and features, their errors tend not to overlap. Averaging (voting) therefore smooths out individual mistakes and boosts overall accuracy. Trees never trained on a given row can still predict that row (these are what are called the out-of-bag cases) so we can estimate test accuracy on the fly without a separate validation set.

[11, 17, 39]

Multinomial Logistic Regression (Softmax Regression) Multiclass logistic regression models the probability of class k given a feature vector $x \in R^d$ by the softmax

$$\Pr(y=k \mid x) = \frac{\exp(\beta_k^\top x)}{\sum_{j=1}^C \exp(\beta_j^\top x)}, \quad k = 1, \dots, C,$$

with parameter vectors $\{\beta_k\}_{k=1}^C$ (one set to 0 for identifiability). The model arises from maximising the conditional log-likelihood

$$\ell(\beta) = \sum_{i=1}^n \left(\beta_{y_i}^\top x_i - \log \sum_{j=1}^C e^{\beta_j^\top x_i} \right),$$

whose gradient is the misclassification residual, and the Hessian is negative semidefinite, ensuring concave optimisation solvable by Newton–Raphson, L-BFGS, or (stochastic) gradient descent. Decision boundaries are linear in feature space, and predicted class is $\hat{y} = \arg \max_k \beta_k^\top x$. Regularisers such as L_2 (ridge) or L_1 (lasso) control overfitting and enable feature selection. [37]

Singular Value Decomposition (SVD) For any $m \times n$ matrix A over R (or C) there exist orthogonal (unitary) matrices $U \in R^{m \times m}$, $V \in R^{n \times n}$ and a diagonal matrix $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$, $\sigma_1 \geq \dots \geq \sigma_r > 0$, such that

$$A = U \Sigma V^\top,$$

where $r = \text{rank}(A)$. The columns of U (left-singular vectors) and V (right-singular vectors) provide orthonormal bases that diagonalise A . Singular values $\sigma_i = \sqrt{\lambda_i(A^\top A)}$ measure the ℓ_2 stretch of the transformation. Truncating to $k < r$ terms yields the best rank- k approximation in Frobenius norm, $A_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$, the backbone of Principal Component Analysis, latent semantic indexing, and reduced-order modelling. [40]

While the models evaluated in this study strike a practical balance between accuracy and speed, the current deep-learning landscape offers even more expressive, albeit heavier, architectures that could push performance further.

Sequence-to-sequence encoder–decoder networks (e.g. Transformers with attention) excel at capturing long-range order in user-interaction histories, allowing the model to "generate" personalised slates of items rather than merely rank a static candidate set. Likewise, Graph Neural Networks (GNNs) propagate information through the user-item interaction graph, extracting high-order collaborative signals and side-relation patterns that traditional methods miss.

These approaches demand larger memory footprints, specialised hardware, and careful batching strategies, yet they represent the state-of-the-art on many public benchmarks. Incorporating them is therefore a promising path for future work.

Encoder-Decoder Architectures An encoder–decoder network maps a variable-length input sequence $(x_1, \dots, x_T)$ to an output sequence $(y_1, \dots, y_{T^*})$ via two learned stages: the encoder converts inputs into a latent representation (context) $\mathbf{c}$, often the final hidden state or an attention-weighted sum of states, while the decoder autoregressively generates outputs conditioned on $\mathbf{c}$ and previously produced tokens. For recurrent implementations,

$$\begin{aligned} h_t &= f_{\text{enc}}(x_t, h_{t-1}); \quad \mathbf{c} = g(\{h_t\}_{t=1}^T), \\ s_{t'} &= f_{\text{dec}}(y_{t'-1}, s_{t'-1}, \mathbf{c}); \quad \Pr(y_{t'} \mid y_{<t'}, \mathbf{c}) = \text{softmax}(W_o s_{t'}), \end{aligned}$$

where $f_{\text{enc}}/f_{\text{dec}}$ may be GRU, LSTM, or Transformer blocks, and g an identity (no attention) or an attention mechanism $g(\cdot) = \sum_t \alpha_t h_t$ with weights $\alpha_t = \text{softmax}_t(s_{t-1}^\top W_a h_t)$. [6, 18]

Graph Neural Networks (GNNs) A Graph Neural Network learns node (or edge/graph-level) representations by iterative message-passing over a graph $G = (V, E)$, generalising convolution to non-Euclidean domains. At layer ℓ each node v aggregates information from its neighbourhood $\mathcal{N}(v)$,

$$h_v^{(\ell+1)} = \sigma(W^{(\ell)} \text{agg}\{h_u^{(\ell)} : u \in \mathcal{N}(v)\} + b^{(\ell)}),$$

with initial features $h_v^{(0)} = x_v$ and non-linear activation σ . Choices of agg (aggregate functions such as mean, sum, max, attention or spectral convolutions) yield specific architectures like GCN, GraphSAGE, GAT, or ChebNet. The permutation-invariant aggregation ensures that the learned function respects graph isomorphisms, and after L layers each node embedding encodes its L -hop receptive field.

A global task (e.g. graph classification) pools node embeddings via $\rho(\{h_v^{(L)}\})$ (mean, sum, set transformer) to produce a graph-level vector. Supervised training minimises loss $\mathcal{L}(y, \hat{y})$ augmented with regularisers (dropout, batch-norm) to mitigate over-smoothing and over-fitting. Theoretical analyses connect these networks to the Weisfeiler–Lehman graph isomorphism test, bounding their discriminative power. [6]

2.4 Evaluation and Distance Metrics

Total Variation Distance (TVD) The total variation distance between two discrete probability distributions P and Q over the same σ -algebra $\mathcal{F}$ quantifies their maximum discrepancy across all measurable events. It is defined as

$$\text{TVD}(P, Q) = \sup_{A \in \mathcal{F}} |P(A) - Q(A)| = \frac{1}{2} \sum_i |p_i - q_i|.$$

The first form shows that TVD is the largest absolute probability gap assignable to any event A , giving it a natural decision-theoretic interpretation: it is the worst-case excess error rate when one uses P in place of Q . The second, ℓ_1 -norm formula arises by decomposing the probability-mass discrepancy into positive and negative parts, and the factor $\frac{1}{2}$ ensures the metric ranges from 0 (identical distributions) to 1 (mutually singular).

TVD induces a metric obeying symmetry, non-negativity, identity of indiscernibles, and the triangle inequality, and it upper-bounds other divergences such as Kullback–Leibler via Pinsker’s inequality. [25, 42]

Jaccard Index Jaccard Index, also known as Intersection over Union (IoU), measures the overlap between two sets, most famously a predicted object region $\hat{B}$ and the ground-truth box B in computer-vision tasks, and is given by

$$\text{IoU}(B, \hat{B}) = \frac{|B \cap \hat{B}|}{|B \cup \hat{B}|},$$

where $|\cdot|$ denotes area (or volume, or pixel count). IoU therefore ranges from 0 (no overlap) to 1 (perfect match) and is thresholded (e.g. $\text{IoU} \geq 0.5$) when computing metrics such as mean Average Precision in object detection. Because both numerator and denominator are scale-dependent, IoU naturally penalises both false positives (expanding the union) and false negatives (shrinking the intersection), making it a balanced set-similarity score that is invariant to identical scaling of inputs. [22, 32]

Jensen–Shannon Divergence (JSD) The Jensen–Shannon divergence symmetrises and smooths the Kullback–Leibler divergence, producing a finite, bounded measure of dissimilarity between two distributions P and Q :

$$\text{JSD}(P \parallel Q) = \frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M), \quad M = \frac{1}{2}(P + Q).$$

Here $D_{\text{KL}}(P \parallel M) = \sum_i p_i \log \frac{p_i}{m_i}$, and the mixture distribution M prevents the divergences from blowing up when either p_i or q_i vanishes. JSD inherits non-negativity, symmetry, and the property that its square root $\sqrt{\text{JSD}}$ is a true metric on the probability simplex, bounded in $[0, \log 2]$ (or $[0, 1]$ if base-2 logarithms are used), which facilitates stable optimisation in applications such as generative modelling and language-model evaluation. [5, 33]

Entropy The (Shannon) entropy of a discrete distribution P quantifies its intrinsic uncertainty or average information content:

$$H(P) = - \sum_i p_i \log p_i.$$

A higher value indicates a more uniform distribution, whereas $H(P) = 0$ for a degenerate (deterministic) distribution. Entropy underlies data-compression limits (source-coding theorem), with $H(P)$ giving the minimal expected code length per symbol in the absence of compression overhead. In statistical inference, the principle of maximum entropy selects the least-committal model subject to known constraints, while in physics the same formula appears in Boltzmann–Gibbs thermodynamics, linking information theory to disorder. For continuous variables one uses the differential entropy $h(f) = - \int f(x) \log f(x) dx$, noting its sensitivity to coordinate transformations and its role in defining mutual information $I(X; Y) = H(X) + H(Y) - H(X, Y)$, which measures the shared information between random variables.

3 Preprocessing

The dataset provided for this thesis consists of university pre-enrolment records for undergraduate degrees offered at public universities in Catalonia. These records are accompanied by comprehensive metadata, both regarding the degree programs themselves and the backgrounds of prospective students. The data spans from 2017 to 2024; however, the analysis has been narrowed down to the most recent years and focuses exclusively on the period from 2021 to 2024 to ensure the relevance and consistency of the findings.

For each academic year, three primary files are available: one containing all pre-enrolment applications, another detailing the assignment of students to university places, and a third serving as the official pre-enrolment register.

The application and assignment files share nearly identical structures, with each row representing an individual student and columns corresponding to various attributes. These attributes include academic records, performance in the university entrance examinations (PAU), and a range of personal, geographic, and family background variables collected during the pre-enrolment process. This structure enables the construction of a detailed and multifaceted profile for each student.

The distinction between the two main datasets lies in their scope: the application dataset encompasses all students who have applied for university admission, while the assignment dataset includes only those who have been allocated a place, with additional fields indicating the assigned and enrolled degree. The assignment process is contingent upon the completion of the PAU or equivalent access routes, after which students are distributed among the available university places.

Students may access university through several pathways, including the standard PAU, previous university degrees, vocational training programmes (CFGS), and special access routes for older students (aged 25, 40, or 45), among others. All data is provided in microdata format, meaning that while direct identifiers such as names or national identification numbers are absent, the combination of a small number of fields could potentially make individuals identifiable. This raises significant ethical considerations, particularly regarding the sensitivity and privacy of the data. To address these concerns, all analyses and presentations in this thesis are conducted on aggregated data, ensuring that no individual can be identified and that their academic integrity is maintained.

The variables available in the datasets are extensive. They include the secondary school attended, student nationality, region of origin (down to the *comarca* or province), year of birth, access route, subjects taken and grades obtained in the PAU or equivalent, academic records, up to eight degree preferences indicated during pre-enrolment, parental education and occupation, and other personal indicators such as elite athlete status or scholarship eligibility. Detailed information about the secondary schools is also available, including their names, postal codes, and whether they are public, private, or semi-private institutions. For students entering through vocational or alternative routes, relevant academic and access information is similarly recorded.

For the purposes of this analysis, the focus has been narrowed to students coming from vocational training (FP) and the baccalaureate (*Batrillerat*) pathways. These groups are the most comparable in terms of data structure and student profiles, and they constitute the majority of new university entrants. This restriction allows for a more precise examination of equity in university access, as these students are typically aged 16 to 24, lack economic independence, and are closely tied to the educational system.

When it comes to the degrees and programmes, a significant filtering process was also applied to exclude specialized master's programs, particularly those designed for future secondary school teachers. These programs constitute a distinct group with different access routes and are not relevant to the analysis of undergraduate access, thus warranting separate consideration.

All data is coded according to the official pre-enrolment register, which provides comprehensive codebooks for all variables. This coding system enhances data robustness by minimizing manual data entry errors and standardizing variable formats. For analysis, a decoding file was created to facilitate the interpretation of results, allowing coded data to be translated into more meaningful categories by mapping them with the encodings found in the register file. This file was also used to enrich the dataset during preprocessing, filling in missing values for school names, postal codes, and other attributes by cross-referencing

available information and performing reverse engineering where necessary. For example, if a postal code or school code was missing, name matching techniques were employed to retrieve the relevant information.

In addition to the original data, external sources were integrated to further enrich the analysis. [12] These sources provided mappings for postal codes and municipalities, as well as socioeconomic indices such as family income levels, drawn from public datasets maintained by the Catalan Institute of Statistics, the tax agency, and the National Statistics Institute. This integration enabled a more nuanced and accurate socioeconomic analysis.

The preprocessing steps resulted in a substantial reduction in the number of records, as shown by the statistics for each year. For example, for the year 2021, the original dataset contained 54,017 records and 32 variables, but after filtering and enrichment, the final dataset comprised 32,074 records and 40 variables. Similar reductions and enhancements were observed across all years analyzed. Despite variations in the number of records, the structure, coding, and variable definitions remained consistent, facilitating the development of replicable and robust analytical procedures and enabling reliable temporal comparisons.

Year	Original shape	Valid Enrolled Center	Valid Secondary School Center Code	Valid Secondary School Center Name	Valid FP Center Code	Valid FP Center Name	Final shape
2021	(54,017, 32)	41,651	34,355	48,282	7,316	8,273	(32,074, 40)
2022	(56,750, 32)	42,659	35,832	50,793	7,542	8,713	(32,350, 40)
2023	(57,324, 32)	43,430	36,327	51,125	8,214	9,339	(31,053, 40)
2024	(57,159, 32)	43,946	35,576	49,540	8,660	9,776	(31,523, 40)

Table 1: Record-Filtering Summary by Academic Year

Ethical considerations are paramount in handling this data. The potential for re-identification, even in the absence of direct identifiers, requires strict adherence to privacy and data protection principles. The data must be used responsibly, and under no circumstances should it be shared or uploaded to public repositories without appropriate consent and safeguards. Manual data completion processes, which are described in detail in a subsequent section, were limited to assigning characteristics and metadata to degree types and did not introduce any bias regarding individual students.

As a final remark, the context of the data is crucial for its correct interpretation. The analysis pertains to Catalonia, an autonomous community within Spain characterized by advanced technological development and a strong tertiary sector. The majority of educational institutions are concentrated in provincial capitals, which may be emphasized in certain parts of the study to illustrate specific findings. While the analysis remains generalizable to the whole of Catalonia, the origins of students are preserved, minimizing bias related to geographic filtering.

Occasional references to specific institutions, such as the GCED, the Degree in Data Science and Engineering at UPC, are included for illustrative purposes and to provide context relevant to the study's setting.

4 Data Analysis: Degree Knowledge Graph

4.1 Initial Approach: Degree Programs as Probability Distributions Based on Application Preferences

With the dataset now thoroughly cleaned and enriched, the next step in the analysis was to understand the demand patterns for university degree programs. The central idea was to model each degree program as a probability distribution, defined by the frequency and order in which it is selected by applicants. This approach allowed for a nuanced examination of how degree programs are chosen, the relative popularity of different programs and the identification of patterns or similarities in applicant behavior across programs and over time.

At this stage, the data consisted of individual student records, each containing up to eight ranked preferences for university degree programs. While some students may list all eight options, others may provide fewer, resulting in a variable number of preferences per applicant. Each preference is encoded as a program code, and the dataset thus represents a collection of ordered lists of degree choices. The primary questions guiding this analysis were: How are degree programs being selected? Are certain programs consistently more in demand than others? What are the observable patterns in application and assignment behaviors? Furthermore, which degree programs exhibit similar demand profiles, and how do these profiles evolve over time?

To address these questions, the preferences for each degree program were aggregated and analyzed as probability distributions. Specifically, for each program, the number of times it was selected in each preference position (first, second, etc.) was counted. These counts were then normalized by the total number of applicants, resulting in a distribution that reflected the relative demand for each program across preference ranks. This method enabled the identification of programs that are frequently chosen as first choices, as well as those that tend to appear more often in lower preference positions.

Let the raw preference counts for a given programme be

$$\mathbf{A} = (A_1, A_2, \dots, A_8) \in \mathbb{N}^8,$$

where A_i denotes the number of students who place the programme at preference level i ($i = 1, \dots, 8$). To compare programmes independently of their absolute popularity, we normalise these counts into a probability mass function

$$\mathbf{a} = (a_1, a_2, \dots, a_8), \quad a_i = \frac{A_i}{\sum_{j=1}^8 A_j}, \quad i = 1, \dots, 8.$$

By construction,

$$a_i \geq 0 \quad \text{and} \quad \sum_{i=1}^8 a_i = 1,$$

so $\mathbf{a}$ lies in the 8-dimensional simplex and can be interpreted as the empirical distribution of preference levels for the programme.

The similarity between the demand distributions of different degree programs was quantified using the Total Variation Distance (TVD). This metric provides a rigorous basis for clustering degree programs with similar demand patterns and for tracking changes in these patterns over time.

Case Study: GCED (Code 31089) and Its Evolving Demand Profile

A specific example of this approach was done by examining the degree program GCED. By analyzing the distribution of preferences for GCED across multiple years, a clear shift was observed. In 2021, GCED was predominantly selected as a second preference by applicants, accounting for 23% of its total selections. However, from 2022 to 2024, GCED increasingly became the first choice, with the proportion of first-preference selections rising to between 21% and 24%. This transition suggests a growing recognition

and desirability of the program, likely driven by recent advances and heightened interest in fields such as artificial intelligence, data science, and related disciplines. The increasing prominence of GCED as a first-choice program reflects broader trends in the labor market and academic landscape, where data-driven and technology-oriented skills are in high demand.

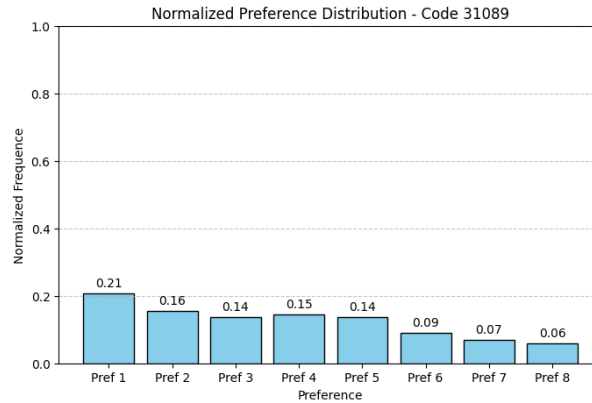


Figure 1: Normalized Preferences Distribution GCED, year 2024

The application of TVD also enabled the identification of other degree programs that share similar demand distributions with GCED in each year. For instance:

- In 2021, programs such as Chemical Engineering, Journalism, Chemistry, Social Education, Tourism (CETT), English and French Studies, Applied Statistics, Electrical Engineering, and joint programs in Business Administration and Mathematics exhibited demanded profiles similar to GCED.
- In 2022, the most similar programs included Automotive Engineering, Primary Education, Business Administration (at Tecnocampus), Electrical Engineering, Telecommunications Engineering, Art History, Marine Sciences and Technologies, ICT Systems Engineering, English and Spanish Studies, and Social Work.
- In 2023, the closest matches were Audiovisual Systems Engineering, Social Education, Marketing and Digital Communication (EAE), Hispanic Philology, Physiotherapy (Gimbernat), Social Education, Product Design and Development Engineering, Labor Relations, English Studies, and Tourism and Hospitality Management.
- In 2024, the programs with the most similar profiles were Artificial Intelligence, Law, Industrial Design Engineering and Product Development, Biotechnology, Chemistry, Sociology, Nursing (Sant Joan de Déu), and Environmental Sciences.

Under the hypothesis that GCED is a niche program, these findings suggest that its demand profile has evolved significantly in recent years. Initially, GCED shared similarities with other specialized or less mainstream programs. However, as its popularity has grown, alongside related fields such as Artificial Intelligence and Environmental Sciences, it has begun to exhibit demand patterns more characteristic of broader, more established programs like Business Administration, Law, Chemistry, Biotechnology, and Nursing. This shift could be a first indication that GCED is transitioning from a niche offering to a more mainstream option, reflecting changing perceptions among prospective students and the increasing relevance of its subject matter in the contemporary academic and professional landscape.

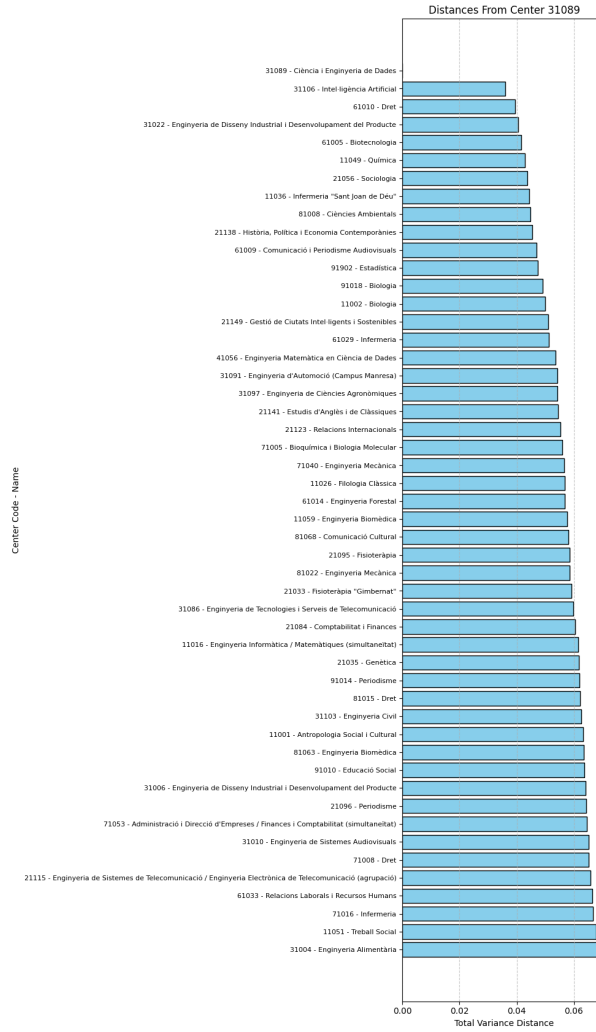


Figure 2: Results of TVD between GCED's and other Degrees' Preference Distributions, year 2024

4.2 Transition to NLP-Inspired Analysis: Frequencies and Co-Occurrences

Given the inherent complexity of analyzing and comparing all degree program demand distributions simultaneously and the challenge of defining a single ideal or threshold distribution, a new analytical perspective was adopted. Instead of focusing solely on the individual distributions of preferences for each degree, the data was reinterpreted through the lens of Natural Language Processing (NLP). This shift allowed for the application of established NLP techniques, such as frequency analysis, co-occurrence measurement, and vector representations, to better capture not only the behavior of individual degree programs but also the intricate web of relationships, similarities, and trends that emerge among the full set of university-level academic programmes.

In this framework, each student's list of degree preferences was treated analogously to a document composed of terms, where each "term" corresponds to a specific degree program. By considering the entire set of student preferences as a corpus, it became possible to ask new questions: Which degree programs are most closely related based on student choices? Do isolated or exclusive communities of programs emerge within the broader academic landscape? Which programs are the most popular, and how do these patterns of popularity and association evolve over time?

Nevertheless, to move beyond simple counts and capture the subtle relationships between degree programs, the concept of a knowledge graph was introduced. In this context, the knowledge graph is a network where nodes represent degree programs and edges represent the strength of association between pairs of programs, as inferred from their co-occurrence patterns in student preferences. The construction of this graph was guided by a carefully defined mathematical formula that incorporates not only the raw

frequency of co-occurrence but also the relative positions of the programs in each student's preference list.

4.3 Initial Knowledge Graph Definition: Parameters and Rationale

The central component of the knowledge graph was the calculation of edge weights, which quantify the strength of the relationship between two degree programs. For each student, all valid pairs of degree programs from their preference list were considered. For each pair, the pairwise positions in which the programs appeared (for example, first and third preference) were also recorded.

To quantify the strength of the connection between any two centres, we assigned a weight

$$w(A, B),$$

defined as

$$w(A, B) = \sum_i \sum_j \left[\frac{1}{|i - j|^\alpha} \times \frac{1}{(i + j - 2)^\beta} \times \frac{\text{co-occurrence}(A_i, B_j)}{\text{Total}_A \times \text{Total}_B} \right],$$

where

- i and j denote the preference positions ($i, j = 1, \dots, 8$ in our data);
- $\alpha > 0$ controls how sharply the weight decays with the difference in positions $|i - j|$;
- $\beta \geq 0$ modulates the decay with the sum of positions ($i + j - 2$), giving extra emphasis to higher-priority preferences;
- $\text{co-occurrence}(A_i, B_j)$ is the number of students who list centre A in position i and centre B in position j ;
- Total_A and Total_B are the total numbers of students listing centres A and B , used for normalisation.

Hence, $w(A, B)$ aggregates all co-occurrences, penalising pairs that are far apart in the preference ranking and down-weighting lower-priority pairs, while remaining scale-free through the normalisation by $\text{Total}_A \times \text{Total}_B$.

α and β are adjustable parameters that control the sensitivity of the model to the difference and sum of the preference positions respectively. The first term, $\frac{1}{|i-j|^\alpha}$, penalizes pairs that are chosen in positions that are far apart, with higher values of α leading to stronger penalties for distant pairs. The second term, $\frac{1}{(i+j-2)^\beta}$, reduces the weight for pairs that are both chosen in lower-priority positions, with higher β values emphasizing the importance of high-priority preferences.

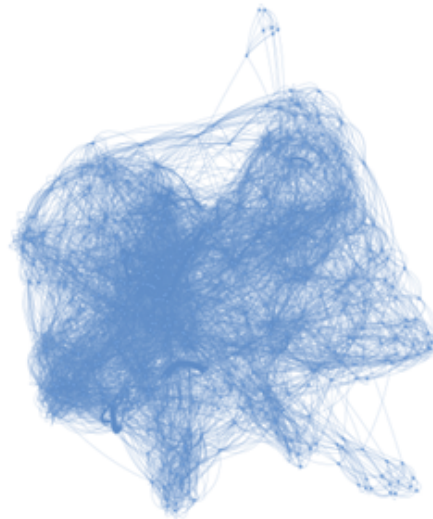
For each pair of degree programs, the total weight is computed by summing the contributions from all students who selected both programs, normalized by the total number of students who selected each program in the relevant positions. This normalization ensured that the resulting weights are comparable across different program pairs and are not excessively influenced by programs with very high or very low overall popularity.

A key advantage of this approach is its flexibility. By adjusting the parameters α and β , as well as a threshold for including edges in the graph, we were able to tailor the sensitivity of the model to different aspects of student behavior. We tested the following values for the parameters:

Test	Alpha	Beta	Threshold	Description
1	0.1	0.1	0.1	Minimal penalty for both difference and position; very dense graph with high weights.
2	0.3	0.3	0.1	Slight increase in penalty; a bit more discrimination is noticeable based on preference proximity.
3	0.7	0.7	0.1	Moderate penalty; weights better reflect proximity-based relationships among student preferences.
4	1	1	0.1	Reference values; clear penalty applied for both difference and overall position priority.
5	1.5	1.5	0.1	Strong penalty; highlights only co-occurrences with very close positions and high priority, resulting in lower weights.
6	1	0.3	0.1	High penalty for position difference but lower for overall position; especially highlights when centers appear together.
7	0.3	1	0.1	Lower penalty for difference but strong penalty for position quality; emphasizes high-priority preferences.
8	Optimized	Optimized	Upper threshold	Final adjustment with best parameters from prior analysis; large threshold filters weak connections and focuses on strong relationships.

Table 2: Parameter Settings and Descriptions

Upon constructing the knowledge graph based on student preferences, several visually distinctive sections emerged, each corresponding to groups of degree programs that are frequently associated in applicants' choices. These regions reflected both disciplinary affinities and broader thematic connections, providing a nuanced map of the academic landscape as perceived by prospective students.

Figure 3: Initial Knowledge Graph with $\alpha = 0.7$, $\beta = 0.7$ and edge threshold = 0.1

For instance, one prominent region encompassed degree programs in electronic engineering, mechanical engineering, mathematics, and computer science. This grouping highlighted the strong interconnections among technical and quantitative disciplines, which are often considered together by students with interests in STEM fields.

Another significant section brought together medicine, pharmacy, biomedical sciences, chemistry, biotechnology, and biomedical engineering. This cluster was organized along a spectrum, with programs ranging from the core health sciences on one end to related engineering disciplines on the other. The structure of this cluster suggested a continuum of interest among students who are drawn to the life sciences and

their technological applications.

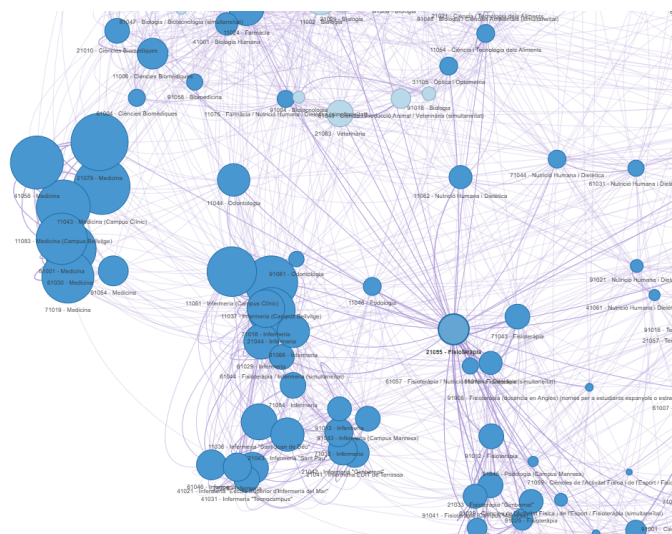


Figure 4: Life Sciences Region of Knowledge Graph

A third region identified was composed of programs in Germanic languages, history, philology, and political science. Extending from this core, the cluster connected to law and criminology, indicating a pathway of related interests that bridge the humanities and social sciences.

Closely related one could identify a region grouping early childhood education, psychology, pedagogy, and primary education. This grouping reflected the close association between educational and psychological disciplines, which are often considered together by students interested in human development and teaching.

A more heterogeneous section emerged, comprising public relations, tourism, marketing, technological business management, architecture, and digital business administration. The central position of this cluster suggested that these programs serve as bridges between more specialized academic communities, attracting students with diverse interests.

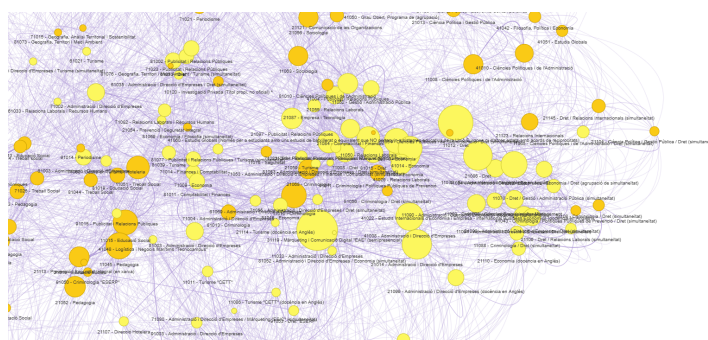


Figure 5: Business and Marketing Region of Knowledge Graph

The structure of the knowledge graph was shaped by the compactness or dispersion of these clusters, the exclusivity of the most strongly connected groups, and the overall density of the network. Compact clusters were characterized by high internal connectivity, indicating that students frequently consider multiple programs within the same cluster as alternatives or complements. In contrast, more dispersed clusters were linked by weaker associations, reflecting broader or less cohesive patterns of interest.

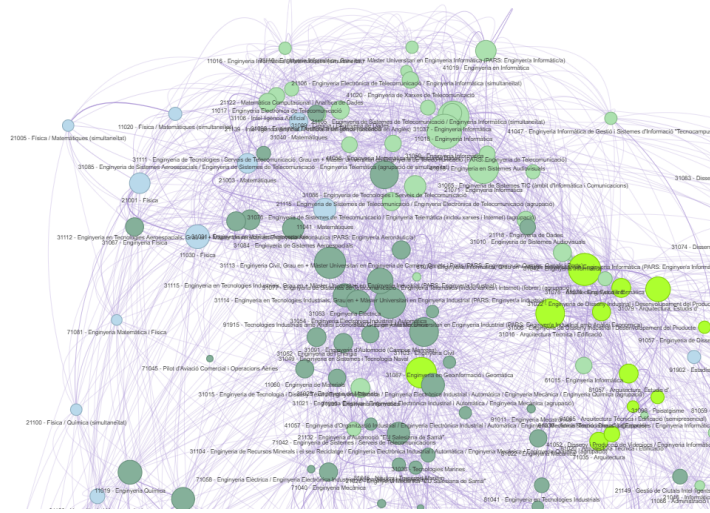


Figure 6: STEM and Pure Sciences Region of Knowledge Graph

During the analysis, it became apparent that the parameter α , which controls the penalty for differences in preference positions, had limited impact on the overall structure of the graph. This is because, in practice, most meaningful co-occurrences occur between programs chosen in similar positions, making the effect of α relatively minor. In contrast, the parameter β , which governs the importance of the absolute positions in the preference list, played a much more significant role. By adjusting β , it was possible to modulate the density of the graph and the relative prominence of different degree programs, with higher values of β emphasizing top-ranked preferences and leading to sparser, more selective networks.

However, the initial definition of the model also revealed other limitations. While the approach successfully captured strong connections between popular and well-known programs, it also tended to highlight strong associations among less popular or more obscure programs. This phenomenon arose because mutually exclusive programs, meaning those that are rarely chosen together with others, can appear highly connected within their own small clusters. The exclusivity of these programs, reflected in their frequent co-occurrence within a limited set of preference lists, artificially inflates the strength of their connections in the graph.

In light of these observations, a second definition of the model was proposed, involving a re-examination and redefinition of the parameters α and β . The goal of this refinement was to achieve a more balanced representation of popular and niche programs, ensuring that the knowledge graph accurately reflected the true structure of student preferences without overemphasizing the exclusivity of less commonly chosen programs.

4.4 Refining the Model: Parameter Adjustments and Enhanced Visualization

The calculation of edge weights was modified to account for the relationship between the number of students choosing both programs and the total volume of selections for those programs.

The weight between centres A and B is now defined as

$$w(A, B) = \sum_i \sum_j w_{ij} \frac{\text{Total}(A_i, B_j)}{(A_i \times B_j)^\beta},$$

with

$$w_{ij} = \frac{1}{(i + j - 2)^\alpha}.$$

- i and j are the preference positions ($1 \leq i, j \leq 8$ in our data).

- $\text{Total}(A_i, B_j)$ counts students who list centre A at position i and centre B at position j .
- $(A_i \times B_j)$ denotes the product of the marginal totals for those two positions (i.e. the number of students who place A in position i times the number who place B in position j).
- $\alpha > 0$ controls the penalty as the *sum* of positions $(i + j - 2)$ increases.
- $\beta \geq 0$ modulates the strength of the normalisation by the marginal totals: $\beta = 0$ yields unnormalised counts, whereas larger β down-weights pairs that are individually popular.

This revised approach introduces two key improvements. First, the normalization by $(A_i \times B_j)^\beta$ allows the influence of the relative size of the programs to be adjusted, ensuring that the weight of an edge is not artificially inflated by a low absolute frequency but is instead considered in relation to the total number of selections for each program. The parameter β , which takes a value between 0 and 1, controls the extent to which this normalization affects the weight of the relationships. This adjustment ensures that the graph more accurately reflects the true strength of connections between programs, regardless of their overall popularity.

Secondly, the weight function was redesigned to depend more on the sum of the preference positions rather than their difference. This means that the impact of the priority given to programs in student preference lists can be fine-tuned using the parameter α . Higher values of α place greater emphasis on the preferences of the top ranked, making the model more sensitive to the most important choices, while lower values distribute the weight more evenly across all positions. This flexibility allows the model to capture both strong, focused preferences and broader, more distributed patterns of interest.

In addition to changes in edge weights, the size of each node in the graph was also redefined to represent the number of times a program has been selected in any preference position. Rather than displaying this value linearly, a scaling function has been introduced to adjust the influence of this value on node size.

Specifically, the node size is now defined as $A^{(1-\beta)}$, where A is the total number of selections for the program and β is a parameter that controls the sensitivity of the scaling. When β is close to 1, differences in node size are less pronounced, resulting in a more uniform visual representation. Conversely, when β is low, programs with many selections appear much larger than those with fewer selections, making popularity visually dominant.

Visualization Enhancements

To further improve the interpretability of the graph, a new layer of information was introduced by assigning each degree program a modality and submodality. This classification was inspired by the structure of high school and baccalaureate academic branches, making the groping system detailed yet manageable. The modalities and submodalities were manually assigned to each program, drawing on established categories to ensure both relevance and clarity.

A color-coding scheme was implemented to visually distinguish between modalities and submodalities. Each main modality was assigned a distinct color gradient, while submodalities within each group were represented by variations of a single color. For example, scientific programs were shaded in blue tones, with specific subfields such as natural sciences, biosciences, and basic sciences each receiving a unique shade. More specifically:

Modality	Submodalities	Color
Scientific	Natural Sciences, Biosanitary Sciences, Basic Sciences (mathematics, physics, chemistry, biology)	Blue
Technological	Basic Engineering and Industrial Engineering, Computer Science and Telecommunications, Industrial Design and Architecture	Green
Social/Humanistic	Economics and Business, Humanities and Social Sciences, Philology and Literature	Yellow
Artistic	Visual Arts and Design, Image and Cinema, Performing Arts, Music, and Dance	Red

Table 3: Modalities, Submodalities, and Assigned Colors

This color-coding system served multiple purposes. It allowed for immediate visual identification of clusters and communities within the graph, making it easier to observe how programs grouped together based on both student preferences and academic classification. It also facilitated more precise filtering and selection of areas for closer inspection.

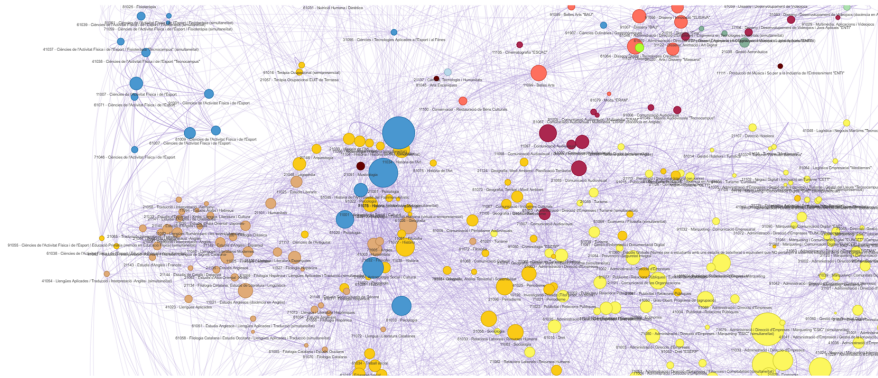


Figure 7: Field Intersection (Psychology - Education) of Knowledge Graph

These improvements resulted in a more balanced and interpretable graph representation.

We tested the following values for the parameters:

Test	Alpha	Beta	Description
1	0.2	0.2	Soft penalty for both distance and global position, yielding more flexible and diverse connections.
2	0.2	0.5	Global position matters more, prioritising higher-ranked choices.
3	0.2	0.8	Strong focus on top-ranked choices; connections between distant options diminish.
4	0.5	0.2	Distance between choices is penalised more than global position.
5	0.5	0.5	Balanced penalties for distance and position, producing moderately restricted connections.
6	0.5	0.8	Strong penalty for distance and emphasis on higher ranks, creating a structured graph.
7	1.0	0.2	Very strong penalty for choice differences; global position has little influence. Distant links are scarce.
8	1.0	0.5	Balanced consideration of distance and hierarchy, retaining some flexibility.
9	1.0	0.8	Maximum penalties for both factors; only close choices connect.

Table 4: Distance and Position Penalty Configuration

Through the testing process, it became clear that an optimal configuration would be one that achieved a more balanced representation, with both α and β set around 0.5. In the visualizations, a threshold was

applied to the edge weights to enhance the clarity of the graph, making it easier to distinguish individual nodes and their connections.

We can see in the following image the resulting graph for the year 2024:

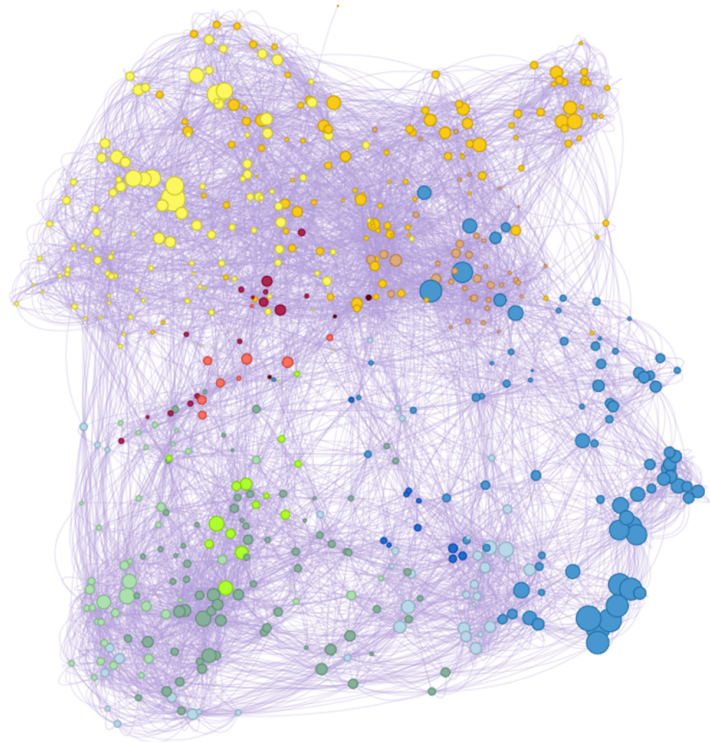


Figure 8: Knowledge Graph 2024

4.5 Temporal Evolution of the Knowledge Graph

The analysis of the knowledge graph over the years revealed significant insights into the evolving structure and dynamics of academic disciplines within the university ecosystem. By examining the knowledge graph manually, it was possible to identify patterns in network formation, the strengthening or merging of clusters, and shifts in the distribution of knowledge and interdisciplinary connections. Nevertheless, it was merely a visual interpretation and hypothetical representation of reality.

4.5.1 Cluster Strengthening and Dispersion: Centrality Measures and Annual Graph Statistics

A crucial step in this investigation involved the calculation of centrality and connectivity measures for each year. While graphical representations of the network provided an initial, intuitive grasp of the relationships between academic programs, a more rigorous, quantitative analysis was necessary to answer the original research questions. Centrality metrics such as degree centrality, closeness centrality, betweenness centrality, eigenvector centrality, and PageRank were computed for each node in the network for every year under study. These measures offer complementary perspectives: degree centrality reflects the immediate connectivity of a node, closeness centrality indicates how efficiently a node can reach others, betweenness centrality highlights nodes that serve as bridges, eigenvector centrality captures influence through connections to other well-connected nodes, and PageRank assesses the overall importance within the network.

The results showed that the distributions of these centrality values remained remarkably consistent across years, typically resembling Gaussian distributions with varying variances but similar ranges. This suggested a stable underlying structure, even as specific nodes rise or fall in prominence.

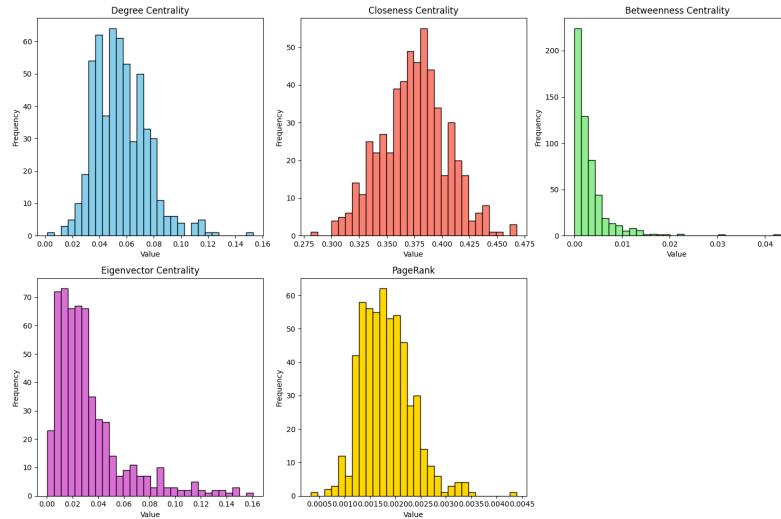


Figure 9: Distributions of the Values of the Centrality

In 2021, the network was characterized by a bicentric core, with Business Administration (ADE) and Psychology occupying the highest positions in degree and closeness centrality. Industrial Technology Engineering and Pharmacy reinforced the STEM and health clusters, while Psychology achieved the highest betweenness centrality, acting as the principal interdisciplinary broker.

By 2022, the dominance of ADE and Psychology persisted, but Mechanical Engineering surged to the forefront in both eigenvector centrality and degree. New entrants such as Sociology (in degree and closeness) and Smart Cities (in betweenness) began to play significant roles. The maximum betweenness centrality slightly decreased, indicating a denser network with more redundant paths and less reliance on single brokers.

In 2023, the leadership in PageRank was shared between ADE and Advertising, while Civil Engineering and a macro-cluster encompassing Textile, Electrical, and Electronic Engineering gained prominence. Psychology maintained its lead in closeness centrality, but its betweenness dropped, suggesting a redistribution of brokerage roles. New brokers, such as Philosophy and Smart Cities, emerged, indicating more distributed pathways across the network.

The year 2024 saw ADE reinforcing its position in both degree centrality and PageRank after a temporary decline. Mechanical Engineering led in eigenvector centrality, and Sociology appeared in several rankings. Smart Cities Management stood out in both closeness and betweenness centrality. The maximum betweenness further decreased, reflecting a more reticulated grid with less dependence on any single bridging node.

Rank	Node	Value
1	Administració i Direcció d'Empreses	0.153005
2	Enginyeria Mecànica	0.123862
3	Enginyeria de Tecnologia i Disseny Tèxtil / Enginyeria Elèctrica / Enginyeria Electrònica Industrial i Automàtica / Enginyeria Mecànica / Enginyeria Química (agrupació)	0.120219
4	Enginyeria Informàtica	0.116576
5	Sociologia	0.116576
6	Enginyeria Civil	0.112933
7	Administració i Direcció d'Empreses	0.112933
8	Psicologia	0.112933
9	Enginyeria en Tecnologies Industrials, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial)	0.111111
10	Farmàcia	0.111111

Table 5: Degree Centrality (Top 10 Nodes) 2024

Rank	Node	Value
1	Psicologia	0.467632
2	Administració i Direcció d'Empreses	0.467234
3	Gestió de Ciutats Intel·ligents i Sostenibles	0.463291
4	Sociologia	0.450000
5	Psicologia	0.448163
6	Ciències	0.442742
7	Economia	0.442742
8	Administració i Direcció d'Empreses	0.442385
9	Finances i Comptabilitat	0.441673
10	Enginyeria de Bioprocessos Alimentaris	0.440964

Table 6: Closeness Centrality (Top 10 Nodes) 2024

Rank	Node	Value
1	Psicologia	0.043123
2	Administració i Direcció d'Empreses	0.030940
3	Gestió de Ciutats Intel·ligents i Sostenibles	0.022352
4	Psicologia	0.021916
5	Enginyeria de Bioprocessos Alimentaris	0.019299
6	Ciències de l'Activitat Física i de l'Esport	0.017652
7	Nutrició Humana i Dietètica	0.017126
8	Sociologia	0.016901
9	Bioinformàtica (docència en Anglès)	0.015030
10	Ciència i Tecnologia dels Aliments	0.013839

Table 7: Betweenness Centrality (Top 10 Nodes) 2024

Rank	Node	Value
1	Administració i Direcció d'Empreses	0.004394
2	Enginyeria Mecànica	0.003506
3	Enginyeria de Tecnologia i Disseny Tèxtil / Enginyeria Elèctrica / Enginyeria Electrònica Industrial i Automàtica / Enginyeria Mecànica / Enginyeria Química (agrupació)	0.003432
4	Sociologia	0.003414
5	Enginyeria Informàtica	0.003369
6	Farmàcia	0.003326
7	Administració i Direcció d'Empreses	0.003295
8	Psicologia	0.003295
9	Enginyeria Civil	0.003236
10	Economia	0.003231

Table 8: PageRank (Top 10 Nodes) 2024

Rank	Node	Value
1	Enginyeria Mecànica	0.160642
2	Enginyeria de Tecnologia i Disseny Tèxtil / Enginyeria Elèctrica / Enginyeria Electrònica Industrial i Automàtica / Enginyeria Mecànica / Enginyeria Química (agrupació)	0.149682
3	Enginyeria en Tecnologies Industrials, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial)	0.147572
4	Enginyeria Civil	0.146453
5	Enginyeria Electrònica Industrial i Automàtica	0.139430
6	Enginyeria Elèctrica	0.138003
7	Enginyeria Informàtica	0.135049
8	Enginyeria de Tecnologies i Serveis de Telecomunicació	0.132700
9	Enginyeria en Tecnologies Industrials, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial)	0.128864
10	Enginyeria de Sistemes de Telecomunicació / Enginyeria Telemàtica (inclou xarxes i Internet) (agrupació)	0.123775

Table 9: Eigenvector Centrality (Top 10 Nodes) 2024

Looking at the global trend from 2021 to 2024, degree centrality remained stably centralized around ADE, but the extreme values of betweenness centrality dropped by 43%, indicating the emergence of more redundant paths and a reduction in unique brokers. The network evolved from a bipolar structure (centered on business and psychology) to a tripolar one, with the addition of a technological-industrial pole (mechanical, civil, and industrial engineering). Engineering clusters shifted from Industrial Technologies in 2021 to Mechanical Engineering in 2023-24, supported by Telecommunications and Electronics.

Emerging degrees such as Smart Cities, Bioinformatics, and Food Innovation entered the network as new brokers, while peripheral fields (arts, philosophy, sports) occasionally became intermediaries, integrating into the core.

It is important to note that centrality is not an intrinsic attribute of a degree but rather the result of its capacity to fulfill complementary needs within the academic ecosystem. For instance, ADE aggregates flows because nearly every professional or research pathway intersects with an economic dimension, while Psychology is pivotal for understanding human factors, serving as a natural bridge to health, education, or marketing. Industrial and mechanical engineering excel when the university requires a common technological platform, reflected in their high eigenvector centrality, as they connect with other influential degrees and amplify their own relevance. Conversely, emerging degrees like Smart Cities exhibit notable betweenness because they fill structural gaps, offering transversal competencies not yet covered by established nodes, thus channeling many new paths through them. Peripheral degrees (arts, philosophy, sports) only achieve peaks in centrality when specific projects, such as digital experience design or sports health, require their expertise, temporarily transforming them into hubs.

4.5.2 Heatmap Representations: Cluster Stability Over Time

We then went on to try a new approach to represent the knowledge graph for a direct, year-on-year comparison of the global structure of the network by plotting it as a heatmap.

To construct the heatmap, we plotted the co-occurrence matrix underlying the knowledge graph. The ordering of rows and columns follows a logical sequence: first by submodality (ranging from natural sciences to the arts), and then, within each group, by ascending program code.

Although some degrees may change codes from year to year, these changes are infrequent and do not significantly affect the overall visualization or the interpretability of the patterns.

A logarithmic color scale was employed to capture the wide range of co-occurrence frequencies. Strongly connected clusters or components appear as lighter colors, progressing from deep purple through pinks to yellows and whites, while low co-occurrence values are rendered as a black background. Because the degrees are ordered by submodality, the most cohesive disciplinary families naturally align along the main

diagonal of the heatmap, forming bright, contiguous blocks.



Figure 10: Log Co-occurrences Heatmap, year 2024

At the top of the heatmap, the natural sciences, biosciences, and foundational disciplines consistently appeared as a compact, nearly white mosaic, reflecting their dense internal connectivity and the prevalence of shared core courses. Immediately following, the large square of basic and industrial engineering was interwoven with computer science and telecommunications, forming an extensive grid. From 2023 onward, increasingly intense vertical traces extended from this engineering block toward the economics cluster, indicating growing interdisciplinary engagement. The industrial design and architecture block, though smaller, was homogeneous and featured bright points where it touched the visual arts, signaling creative cross-occurrence. The economics and business rectangle stood out as the focal point outside the diagonal: its vertical and horizontal bands traversed nearly the entire matrix, visually confirming the role of Business Administration (ADE) as a global hub. Below, the humanities and social sciences, together with philologies, maintained a medium intensity, enabling connections both with the biosciences (via psychology) and with the arts (through communication and literature). At the lower end, the visual arts, film, and performing arts formed a clear but less dense triangle, which gradually gained points of contact with architecture and, from 2023, with creative engineering fields linked to video games.

The evolution of the heatmap is closely connected to the centrality measures previously discussed. In 2021, the map is distinctly modular: the diagonal is sharply defined, and the dark background indicates limited interdisciplinarity, regulated by a few key bridging degrees, most notably Psychology, which that year exhibits the highest betweenness centrality. In 2022, initial whitening appears in the area connecting engineering and economics, and more pink pixels emerge between the humanities and biosciences. The system begins to densify, thanks to the introduction of double degrees and transversal projects such as Smart Cities or Applied Sociology, with the blocks extending toward the margins. By 2023, numerous scattered spots appear outside the diagonal, especially within the engineering, computer science and economics triangle, revealing the entry of mixed degrees like Advertising and Marketing or Civil Engineering with a sustainability focus. A new rectangle also emerges between basic sciences and visual arts. In 2024, the background acquires a finer grain, an economic band traverses the entire matrix toward the arts, and, in the opposite direction, Smart Cities projects a bridge toward the sciences. The network no longer consists of isolated blocks, confirming the emergence of a tripolar structure.

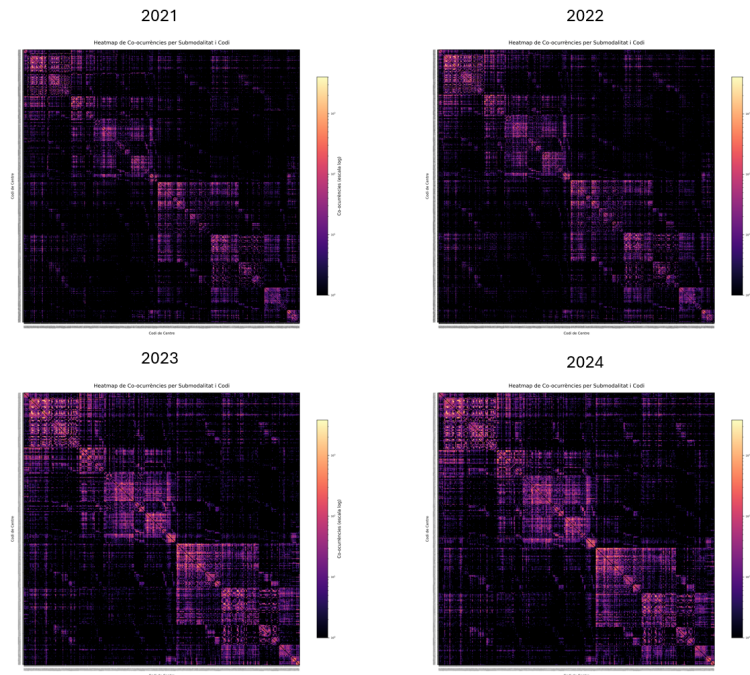


Figure 11: Heatmap Evolution, years 2021-2024

4.5.3 Top Co-occurrences: Identifying Persistent Niches

To further explain which academic degrees exhibit the highest levels of co-occurrence within the knowledge graph, we ranked all possible degree pairs by their frequency of co-occurrence, extracting the top 100 most frequent pairs for each year. This approach allowed us to find the most significant connections from a vast array of possible combinations, providing a clear, data-driven confirmation of the patterns observed in the heatmap and network analyses.

In 2021, the ranking is dominated by the biosanitary cluster. The top twenty-two positions are exclusively occupied by combinations of Medicine degrees from different campuses (such as Clínic, Bellvitge, and Sant Pau), with co-occurrence values ranging from 4,035 to 3,224. This concentration demonstrates the existence of a hyper-dense core within the biosanitary domain. Only further down the ranking do we begin to see the emergence of bridging degrees: repetitions of Psychology with itself at other campuses or universities (UB - UAB), with 2,497 co-occurrences, validates its role as an interface between health and social sciences. Nursing pairs (UB Clínic - UB Bellvitge and UB Bellvitge - UAB), with values between 1,967 and 1,775, form a peripheral ring around Medicine, while Business Administration (with counts equaling 1,618 for UB and UAB) and ADE-Economics (1,141) at closely related centers (UB - UB) already signal the presence of a secondary economic pole, though still largely disconnected from the biosanitary core. Other degrees such as Law (UB - UAB with 1,486), Education at the main universities in Barcelona (UB - UAB) (1,072), and the first STEM indicators like Computer Engineering (UB - UPC, 974) are present, but all remain clearly subordinate to the medical hegemony.

The year 2022 largely repeats this pattern, but with an increase in absolute values (the maximum rises to 4,081 for UB Clínic - UAB) and, more importantly, a greater diversity in the lower half of the top-100. Psychology pairs increase to 2,608 co-occurrences for UB-UAB, Nursing (Bellvitge - Clínic) surpasses 2,200, and ADE-Economics pairs approach 1,416 (UB-UB), nearly 300 more than the previous year. At this point, the network between Computer Engineering (pair 11018 (UB) - 31059 (UPC), with 1,111 co-occurrences) and economic degrees becomes statistically visible. This numerical evidence confirms the whitening observed in the heatmap, which connected the engineering block with the economic band, while the repeated presence of double Nursing degrees, with values between 1,700 (UB Clínic - UAB) and 1,100 (UB Bellvitge - UAB Sant Pau), reflects the densification of the biosanitary belt.

In 2023, the absolute cadence of co-occurrences decreases slightly (the peak drops to 3,839 for medicine in

UB Clínic - UAB), but the composition of the ranking becomes more balanced. Medicine pairs continue to occupy the top positions, but in the value range between 1,100 and 800, consolidated transversal pairs proliferate. Psychology pairs remain at 2,455 (UB - UAB), but ADE-Economics (1,368 for UB - UB) and ADE with itself (UB - UAB with 1,577) gain prominence. New bridges also appear, such as art-tech and architectural pairs (Technical Architecture-Architecture in UPC with 720 co-occurrences) which were absent in 2021.

By 2024, maximum values rise again to 3,849 for Medicine in UB Clínic - UAB, but the distribution is now distinctly tripolar. The biosanitary block remains at the forefront (thirteen of the top fifteen pairs are still variants of medicine) but the economic pole consolidates, with ADE-Economics (1,629 co-occurrences for UB - UB) and ADE pairs (with 1,724 for UB - UAB) both surpassing the 1,600 mark for the first time. The third pole, representing behavior and humanities, is reflected in Psychology pairs (UB - UAB with 2,450) and the sustained appearance of Pedagogy pairs (UB - UAB, 773) and Early Childhood Education-Primary Education in UB and UAB (1,186). Simultaneously, the technological axis asserts itself with Computer Engineering (1,045 for UB and UPC) and Computer Engineering-Informatics pairs (UB - UPC, 886), now firmly within the top-40. At the lower end of the ranking, new "gateways" begin to emerge: Architecture pairs in Barcelona (UPC, 951), Chemistry with Chemistry at other nearby campuses (UB - UAB with 728 co-occurrences), and Physiotherapy EUSES at UdG and Physiotherapy in UVic (674), all corroborating the ongoing expansion of the network.

Case Study: GCED

Focusing on a more detailed case, the degree in Data Science and Engineering (GCED) offered a nuanced perspective on specialization and interdisciplinary expansion. Although it remains firmly rooted in its original "computational STEM" ecosystem, its orbit has gradually expanded toward other quantitative engineering fields, without decisively crossing into the social sciences or health domains.

In 2021, the degree's connections were almost monolithic: thirteen of the top twenty links are with identical or closely related degrees within its submodality (Computer Engineering, Data Engineering, Artificial Intelligence), while the remainder were distributed among Mathematics/Physics (basic sciences) and a handful of industrial or aerospace engineering degrees. Connections with other domains were minimal, with only a single pair involving Medicine and a few double degrees combining business and mathematics.

By 2022, the absolute number of co-occurrences increased. While the same three satellites (Computer Engineering, Mathematical Engineering in Data Science, and Data Engineering) continued to dominate, a belt of industrial engineering degrees was established, and, in the middle of the ranking, integrated degrees in economic analysis or civil engineering began to appear. This shift signals the degree's initial forays into projects with productive applications and operations management.

In 2023, as the peak co-occurrence value decreased slightly and the composition of connections diversified further. Links with Telecommunications, Electronics, and Physics remained strong, while the Mechanical/Aerospace Engineering family entered the top-30 in a stable manner. For the first time, Architecture and Industrial Design also appeared, indicating growing permeability toward areas requiring complex data modeling, though not strictly within the cluster of computer science.

In 2024, the maximum number of co-occurrences dropped significantly, but the network of connections became structurally more heterogeneous. In the lower half of the ranking, degrees such as Bioinformatics, Biomedical Engineering, and Economics appeared with still modest numbers, but their periodic presence confirms a gradual expansion beyond the purely ICT domain.

To answer the central question, the degree in Data Science and Engineering has remained a highly specialized, niche program. However, its evolution reveals a gradual shift from a very tight concentration (with 80% of links within its own cluster in 2021) to a more open matrix, where nearly a third of its relationships now involve industrial, aerospace, biomedical engineering, or economic applications. While the degree does not "generalize" in the sense of forming strong associations with the humanities or strictly health-related fields, it is increasingly positioned as a key node within a broader STEM subset, reflecting both its specialized nature and its growing interdisciplinary reach.

Rank	Code	University	Name (Pair)	Submod.	Count
1	31059	UPC	Enginyeria Informàtica	5	156
2	41056	UPF	Enginyeria Matemàtica en Ciència de Dades	5	138
3	21118	UAB	Enginyeria de Dades	5	123
4	31106	UPC	Intel·ligència Artificial	5	122
5	31040	UPC	Matemàtiques	3	96
6	31115	UPC	Enginyeria en Tecnologies Industrials, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial)	4	81
7	21122	UAB	Matemàtica Computacional i Analítica de Dades	5	79
8	11041	UB	Matemàtiques	3	76
9	11018	UB	Enginyeria Informàtica	5	73
10	31086	UPC	Enginyeria de Tecnologies i Serveis de Telecomunicació	4	69
11	31067	UPC	Enginyeria Física	4	55
12	41019	UPF	Enginyeria en Informàtica	5	54
13	31116	UPC	Enginyeria Informàtica, Grau en + Màster Universitari en Enginyeria Informàtica (PARS: Enginyer/a Informàtic/a)	5	52
14	21071	UAB	Enginyeria Informàtica	5	47
15	21139	UAB	Intel·ligència Artificial / Artificial Intelligence (docència en Anglès)	5	44
16	21003	UAB	Matemàtiques	3	42
17	91915	UPC/UPF	Tecnologies Industrials amb Anàlisi Econòmica, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial amb Anàlisi Econòmica) (docència en Anglès)	4	40
18	31099	UPC	Enginyeria Electrònica de Telecomunicació	4	39
19	11016	UB	Enginyeria Informàtica / Matemàtiques (simultaneïtat)	5	34
20	31112	UPC	Enginyeria en Tecnologies Aeroespacials, Grau en + Màster Universitari en Enginyeria Aeronàutica (PARS: Enginyer/a Aeronàutic/a)	4	33

Table 10: Top 20 Co-occurrences with 31089, Ciència i Enginyeria de Dades, UPC, year 2024

4.6 Clustering the Knowledge Graph

4.6.1 Community Detection and Clustering Methods

As mentioned earlier, to further analyze the structure of academic degree relationships within the knowledge graph, we began by manually assigning modalities and submodalities to each university degree. This classification was based on the content and official description of each program, aiming to reflect the disciplinary reality as accurately as possible. However, a key question arose: how well does this manual categorization align with the actual patterns of co-occurrence between degrees and the emergent structure of the degree network? To address this, we explored whether algorithmic community detection and clustering methods could yield a more accurate or meaningful classification.

Our investigation proceeded in two main stages. First, we allowed community detection algorithms to autonomously determine the optimal number of divisions within the network, thereby revealing the natural groupings present in the data. Subsequently, we applied clustering algorithms with various fixed numbers of clusters to assess whether modularity (the measure of the strength of division of a network into modules) could be improved by adjusting the granularity of the partitioning.

In parallel, we leveraged an independent manual categorization provided by the Academic Management Service (SGA) of UPC, which labeled the fields and subfields (analogous to modalities and submodalities) of UPC's own degrees. This classification served as a benchmark against which we could compare the results of our algorithmic approaches, with the aim of proposing a potentially improved taxonomy.

Our analysis focused on several key aspects:

- The number of clusters generated by each method
- The size distribution of the resulting clusters
- The thematic coherence of the degrees grouped within each cluster

We employed several state-of-the-art community detection algorithms, each with distinct mechanisms for identifying clusters within the network:

- Label Propagation Communities
- Greedy Modularity Optimization
- Louvain Communities

To complement the community detection approaches, we also applied clustering algorithms that require the number of clusters to be specified in advance:

- Spectral Clustering (Laplacian + K-means)
- Agglomerative Clustering
- Hierarchical Clustering

The results can be summarized into the following ideas:

Size	Cluster Label
76	Biological Sciences-Life Sciences
310	Humanities-Social Sciences-Business-Arts
127	STEM-Engineering
12	Nursing
25	Education

Table 11: Cluster Summary – Label Propagation

- Clusters 1, 4, and 5 are thematically very pure (Biological Sciences-Life Sciences, Nursing, Education).

- Cluster 3 successfully groups the technological-industrial domain and exact sciences.
- Cluster 2 acts as an overly broad container. Modularity could improve by exploring more partitions (e.g., splitting Arts/Design, Tourism-Sports, Business-Economy, Social Sciences-Legal, etc.).

Size	Cluster Label
264	Social Sciences-Business-Humanities-Arts
246	STEM-Biological Sciences-Health
25	Education
15	Creative & Games Design

Table 12: Cluster Summary – Greedy Modularity

- Clusters 3 and 4 are thematically clean, confirming that Education and the "arts-design-games" ecosystem have distinct interaction profiles.
- Cluster 1 reproduces the issue seen in Label Propagation: the algorithm merges arts, business, and social sciences into one block due to shared double degrees and cross-disciplinary courses. Finer partitioning is needed for clearer categories.
- Cluster 2 shows the limit of the "maximum modularity" criterion: Engineering and Biosanitary fields co-occur often (e.g., biomedical engineering, bioinformatics), which keeps them together, but for disciplinary clarity, they should ideally be separated.

Size	Community Label
73	Bio-Life & Environmental Sciences
43	Media-Design-Games
80	Humanities & Languages
74	Business, Law, Politics
114	STEM Engineering & ICT
66	Health Care & Sport
54	Marketing, Tourism & Digital Business
46	Education & Social Care

Table 13: Cluster Summary - Louvain Communities

- High adequacy: Louvain produces clean thematic clusters in Bio-Life (1), Media-Design (2), Education-Social (8), and Health-Sport (6).
- Refined division in the social area: Business-Law-Politics (4) separates from Marketing/Tourism (7) and from Humanities (3), improving granularity compared to Greedy.
- Compact STEM cluster: all engineering and exact sciences converge in cluster 5, reflecting their strong cross-co-occurrence pattern.
- Minor inconsistencies (e.g., Integral Safety and Prevention, ESERP Criminology) are due to double degrees or shared campuses but don't distort the main logic.
- Overall, Louvain's automatic labeling aligns well with our manual classification and even refines it in the business-marketing-tourism area. This suggests we could adopt these eight blocks as a new proposal for submodalities, or subdivide mega-clusters 1 (Bio-Life) and 5 (STEM) into two sub-blocks each, to better capture differences between natural and biomedical sciences or between ICT and industrial engineering.

Size	Cluster Label
47	Humanities & Languages
37	Human-Soc Mixed
42	Media-Design-Video Games
60	Mixed Miscellanea
37	Education & Social Care
50	Marketing–Tourism–Digital Biz
58	Business–Law–Politics
45	ICT–Math–Aero–Data
35	Sport–Physio–Nutrition
53	Industrial Engineering & Territory
56	Bio–Life Sciences
30	Medicine & Nursing

Table 14: Cluster Summary – Spectral Clustering

- The 12-cluster configuration offers strong thematic refinement across creative, education, sport-health, and biomedical fields.
- Cluster 11 (Bio-Life Sciences) and 12 (Medicine-Nursing) are clearly separated, reflecting the distinct academic and professional tracks within the health domain.
- Cluster 6 (Marketing-Tourism-Digital Business) is well-defined and differentiates itself from Business, Law and Politics (Cluster 7), which focuses more on policy and corporate disciplines.
- Clusters 2 and 4 are thematically hybrid: Cluster 2 blends humanities, tourism, and business; Cluster 4 gathers "bridge degrees" like Business-Law and Engineering-Business combinations. These likely represent high cross-disciplinarity or algorithmic limits in isolating mixed profiles.
- The structure captures functional logic well: Bio-Life (11), Clinical Health (12), Sport-Health (9) form a coherent health gradient, while ICT–Math (8), Industrial Engineering (10) form the tech-industrial axis. Business (7) and Marketing and Tourism (6) show clear economic sector separation.
- These results suggest the spectral method balances modularity and interpretability, though Cluster 2 and 4 could benefit from additional subdivision to resolve internal diversity.

Size	Cluster Label
30	Sport & Applied Health
17	Tourism & Global Studies
86	Humanities-Communication-Languages
99	Broad STEM & Technology
70	Life & Environmental Sciences
72	Business-Law-Politics
23	Creative Arts & Design
18	Hybrid "Bridge" Cluster
34	Clinical Health
46	Marketing–Tourism–Digital Business
7	Security & Prevention
48	Education–Psychology–Social

Table 15: Cluster Summary – Agglomerative Clustering

- Produces clusters very similar to Spectral for key domains like health (9), bio-environmental sciences (5), education (12), and the arts (7).
- Combines all of STEM into a single giant cluster (4) and does the same for a broad humanities cluster (3), suggesting that average distance within those domains is low; finer modularity might be gained by splitting them.

- Clusters 2 and 8 serve as “catch-all” groups for highly interdisciplinary double degrees, a useful indicator of where to refine manual tagging.
- Overall, the algorithm confirms the robustness of major academic domains, while showing that mixed degrees (e.g., Business-Engineering, Tourism-Advertising, Psychology-Sport) tend to blur strict boundaries.

Size	Cluster Label
24	Tourism, Global Studies & Communication
57	Business-Marketing-Creative Design
7	Food Science & Environment
52	Communication-Public Management-Business
67	Humanities & Languages
25	Early Education
61	Social Welfare & Criminology
34	Creative Tech & Video Games
73	Hard Engineering & ICT
61	Applied Health & Sport
32	Chemical-Bio-Environmental Engineering
57	Life Sciences & Medicine

Table 16: Cluster Summary – Hierarchical Clustering

- Confirms the existence of the major academic blocks: Education, Humanities, Business, Engineering, Health, and Life Sciences.
- Clusters 2, 4, and 8 reveal increasing hybridization between business, marketing, and creative/tech sectors.
- Cluster 3 is a loosely grouped “miscellaneous” cluster, suggesting that even with $k = 12$, some topics remain diffuse.
- Further refinement (e.g., through sub-clustering or adjusting similarity weights) could improve modularity and thematic clarity.

Modularity Comparison Across Clustering Methods

A central metric in evaluating the quality of these partitions is the *normalized modularity* (Q), which quantifies the extent to which a given division of the network concentrates more edges within communities than would be expected in a random graph with the same degree distribution. A value of Q approaching 1 indicates highly cohesive, well-separated blocks, while Q near 0 suggests that the partition offers little improvement over random assignment.

The results for our co-occurrence graph are as follows:

- **Spectral Clustering:** Normalized modularity $Q \approx 0.5306$
- **Agglomerative Clustering:** Normalized modularity $Q \approx 0.4872$
- **Hierarchical Clustering:** Normalized modularity $Q \approx 0.4267$

The spectral clustering approach achieved the highest modularity ($Q \approx 0.53$), despite not explicitly optimizing for modularity. This is because the method projects degrees onto the leading eigenvectors, which naturally delineate the least connected regions of the network, and K-means then partitions the network at its most porous boundaries. As a result, spectral clustering effectively recovers the partition with the highest internal density.

Agglomerative clustering, using precomputed distances, yielded a lower modularity ($Q \approx 0.49$). While it initially merges very similar nodes and forms coherent communities, as the process continues and clusters are forced to merge up to the imposed cut (e.g., $k = 12$), it inevitably combines groups with few real

connections. The greedy nature of the algorithm, where merges cannot be undone, means that early decisions can negatively impact overall cohesion.

Hierarchical clustering, when the dendrogram is cut to produce twelve clusters, resulted in the lowest modularity ($Q \approx 0.43$). Here, the problem is twofold: the merging process inherits even more forced early decisions, and the cut point is determined by the desired number of clusters rather than the point of maximum modularity.

The ordering $Q_{\text{spectral}} > Q_{\text{agglomerative}} > Q_{\text{hierarchical}}$ clearly reflects the degree to which each algorithm attends to the global structure of the network (with spectral clustering offering a holistic perspective) versus making local, irreversible decisions (as in the hierarchical methods). A modularity value of 0.53 confirms that the graph contains well-defined natural blocks and that the spectral partition is the strongest candidate for proposing a new, empirically grounded taxonomy of degrees. The other methods serve as useful benchmarks, illustrating that as the objective of maximizing modularity stops being the priority, the quality of the clustering correspondingly diminishes.

4.6.2 Benchmarking Against UPC's Classification

When analyzing the classification of university degrees provided by UPC, which closely mirrors the modality and submodality framework previously established, a central question emerges: can this existing categorization be improved through algorithmic methods? To address this, we applied both community detection algorithms and clustering methods with a fixed number of clusters to the degree co-occurrence graph, and evaluated the resulting partitions using modularity as a key metric.

The application of these algorithms to our graph presents a significant analytical challenge, given the sheer diversity and number of degrees involved. However, by focusing on a more contained context, specifically, the UPC's own degree offerings, we were able to observe the effects of these methods in a more controlled and interpretable setting.

Modularity Optimization with Greedy Modularity

One of the primary approaches employed was the Greedy Modularity algorithm. Using UPC's reference labeling scheme, the modularity achieved was approximately $Q \approx 0.36$. In contrast, the partition generated by the Greedy Modularity algorithm reached $Q \approx 0.48$, representing a substantial relative increase. This improvement is primarily attributable to the formation of highly compact clusters, specifically clusters 0, 1, 3, 4, and 5, which exhibit strong internal cohesion.

However, a notable weakness remains in Cluster 2, which aggregates too many loosely connected disciplines. This lack of dense internal connections dilutes the overall modularity and highlights the limitations of the current partition.

Cluster	Size	Thematic Summary	Observations
0	20	Pure STEM cluster oriented toward ICT, data science, telecommunications, and aerospace, supported by strong theoretical foundations in mathematics, physics, and statistics.	High cohesion: all degrees involve intensive computation and information engineering. The only outlier is Economics/Statistics (in a double degree), but its relevance to data analytics makes it acceptable.
1	16	Core Industrial Engineering: mechanical, electrical, electronic, materials, energy, chemical, plus PARS industrial engineering tracks.	Very coherent. Biomedical Engineering could be seen as slightly divergent, but its engineering-focused side (devices) justifies its inclusion.
2	13	Miscellanea: maritime/naval, agri-food, environmental, optics, and some multimedia/telecom tracks.	Too heterogeneous: combines naval/marine engineering, ocean sciences, agri-bio, gastronomy, optics/optometry, and even a telematics program. A split is recommended (e.g., marine/naval vs. agri-bio vs. special ICT).
3	10	Private-sector business and marketing programs from EUNCET / EAE (on-site and online business admin, digital marketing).	Very high coherence and clearly distinguishable; no inconsistencies.
4	9	Built environment and territory: architecture, civil engineering, industrial design, geoinformation, and landscape studies.	Consistent; all degrees involve physical space design and management.
5	5	Video games, multimedia, and digital arts (mainly Terrassa and Barcelona campuses).	Small but very clean cluster. Could be merged with cluster 0 for a broader ICT-creative group, but it provides useful granularity on its own.

Table 17: Cluster Summary – Greedy Modularity Results

Several refinements could further enhance both the modularity and interpretability of the clustering:

- Splitting Cluster 2: This heterogeneous cluster could be divided into more coherent subgroups:
 - Marine-Naval: Nautical, marine, and naval engineering degrees.
 - Agro-Food: Agricultural, food science, and gastronomy programs.
 - Optics and Bioinformatics: These could be reassigned to Health or ICT clusters, respectively.
- Repositioning Biomedical Engineering: Evaluating whether Biomedical Engineering would better fit within a future Health-Technology cluster, along with Optics and Bioinformatics, could improve thematic coherence.
- Grouping Multimedia and Video Games: Integrating all multimedia and video game degrees (currently in Cluster 5) with the ICT axis (Cluster 0) may be beneficial if the aim is to form broader, functionally unified blocks. However, retaining their independence can also preserve clearer sector-specific distinctions.

Implementing these refinements is likely to push the modularity beyond $Q \approx 0.52$, while simultaneously reducing the number of clusters and increasing the interpretability of the resulting categories.

Modularity Optimization with Spectral Clustering

When applying spectral clustering, we achieved a normalized modularity of $Q = 0.436$ at its optimal configuration. This value is higher than that obtained for any other tested number of clusters (k). The evolution of modularity as k varied provided valuable insights into the structure of the degree network.

Cluster	Size	Thematic Summary	Observations
0	5	EAE – Business and Digital Marketing (ADE and Marketing, in-person, blended, and English tracks)	Very high cohesion.
1	14	Industrial Engineering - Mechanical, electrical, chemical, energy, materials engineering and ICT systems	Very robust, though ICT systems engineering might better fit in cluster 2.
2	9	ICT - Data - Telecoms, AI, data science, computer science, mathematics	Strong cohesion. Mathematics supports theoretical foundation.
3	8	Construction and Territory - Civil engineering, infrastructure, architecture, landscape, industrial design	Coherent. Includes a creative angle (e.g., product design).
4	8	Bio - Environmental - Marine sciences, agriculture, environmental, food science, biomedicine, optics, bioinformatics	Too broad: optics and biomedicine could form a distinct health-related subcluster.
5	5	EUNCET - Business and Digital Marketing (ADE + Marketing, online)	Analogous to cluster 0, but based in Terrassa; completely clean.
6	5	Network Telecommunications – Systems engineering, telematics, audiovisual communication	Very clean. All share a common infrastructure communication focus.
7	4	Maritime - Naval - Nautical engineering, naval architecture, marine geoinformation	Compact and coherent.
8	9	Advanced Aero - Industrial - Aerospace, industrial engineering with economics, physics, statistics	Includes two sub-blocks (aerospace vs. industrial analysis). Still acceptable.
9	4	Video Games and Multimedia - Design, digital art, multimedia	Clear and distinctive.
10	2	Sport and Gastronomy (FitnessTech and Culinary Arts)	Very small, sparse cluster. Negatively affects modularity.

Table 18: Cluster Summary - Spectral Clustering (Best Partition, $k = 11$)

With $k = 12$, the normalized modularity was $Q = 0.426$. At this level, the algorithm maintained a separation between two ICT subgroups (one focused on network technologies and another on data science) while areas such as sports and gastronomy remained embedded within a broader bio-environmental macrocluster. Additionally, the presence of very small clusters, such as those for video games and marketing, reduced overall internal density.

Reducing to $k = 11$ corrected several of these issues: sports and culinary studies formed their own distinct cluster, and certain degrees, such as Biomedical Engineering, were reallocated to the bio-environmental block. This reorganization resulted in more compact groups and raised modularity to its observed maximum of $Q = 0.436$.

At $k = 10$, the sports-culinary cluster was reabsorbed into the ICT-Telecommunications group, and a small nucleus of science and statistics became isolated, forming a heterogeneous group of twelve degrees. This weakened thematic coherence and caused modularity to drop slightly to $Q = 0.430$.

With $k = 9$, the situation deteriorated further: the maritime-naval block merged with statistics and geoinformation, while industrial engineering absorbed biomedical studies, resulting in an overly broad macrocluster.

The main gain from reducing k from 12 to 11 came from separating an overly diverse conglomerate. Beyond this point, any further merging forced the combination of disparate domains, and when the large ICT cluster absorbed unrelated areas (such as sports, culinary, or maritime studies), the quality of the community structure declined immediately. Thus, $k = 11$ offers the best balance between the number of groups and internal density.



Figure 12: Clustering Benchmarking UPC Degrees

5 Data Analysis: Flows and Socioeconomic Patterns

Having established a comprehensive understanding of how academic centers and degree programs are interconnected, both in terms of thematic content and demand profiles, we found it was essential we delve deeper into the underlying patterns that shape these relationships. Specifically, it was crucial to investigate the socioeconomic factors that influence the profiles observed within various degree programs. This line of inquiry raised several fundamental questions: What socioeconomic patterns underlie the distribution of students across different degrees? From which backgrounds do students originate? Is access to higher education equitable, or do certain groups face barriers? Are there degree programs that remain largely unknown or inaccessible to specific segments of the population due to a lack of information or awareness about emerging or specialized fields? Ultimately, could the root of these disparities be more deeply embedded in the social fabric, making them particularly challenging to address?

To approach these questions, two complementary analytical strategies were employed. The first involved identifying which variables were most strongly associated with our target variable: the academic center to which students are ultimately assigned. The second strategy entailed a more granular mapping of students' postal codes to external factors such as average family income, thereby enabling a clearer and more precise understanding of the socioeconomic flows between pre-university educational institutions and university degree programs.

5.1 Variable Analysis Relative to Assigned Center

5.1.1 Principal Component Analysis

A principal component analysis (PCA) was conducted to transform the complex, multidimensional data into a set of components that capture the most significant sources of variance. This approach allowed for the identification of the variables that most strongly influence the assignment of students to specific academic centers.

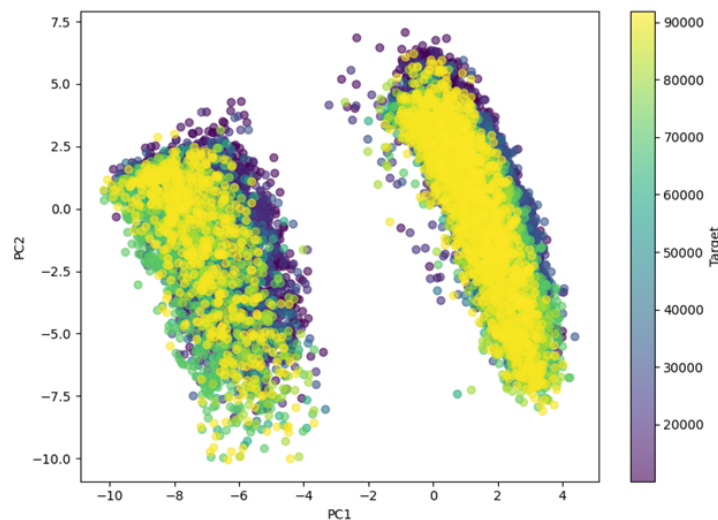


Figure 13: First Principal Component

The first principal component (PC1), which accounts for the largest share of variance, was predominantly shaped by variables related to the students' access pathways and the codes of their previous educational centers. The top ten variables contributing to PC1 included access routes, the codes and frequencies of vocational training centers, and indicators of whether these centers are public. This clear differentiation between students coming from the baccalaureate track and those from vocational training suggested that the type of prior education plays a decisive role in shaping subsequent academic trajectories. It is likely that even more distinct clusters would emerge if additional student categories, such as those transferring from other degrees or older students over 25, were included in the analysis. The data reveal that not only do access routes and center codes matter, but the public or private status of the institutions also

emerged as a significant factor. This observation brings us back to the initial question: To what extent does a student's background condition their entry into university? While a definitive answer remains undetermined, it is evident that these factors have a substantial influence on the data and the observed patterns.

The second principal component (PC2) is defined primarily by variables related to students' stated preferences, particularly the order in which they rank their choices. The top variables here included the frequencies and values of preferences. This component highlighted the importance of individual choice and priority in the assignment process, suggesting that the way students rank their options can create distinct boundaries between different groups.

When examining the components that explain the greatest proportion of variability in the target variable (assigned center), the following results were obtained: PC3 (0.701874), PC5 (0.191021), PC8 (0.122231), PC4 (0.089692), and PC9 (0.086855).

PC3, in particular, is defined by the first three preferences, indicating that students often succeed in enrolling in their desired programs or, at the very least, in accessing their preferred fields of study. This component is also influenced by the students' postal codes, suggesting that geographic location plays a significant role in the selection of academic centers. This may be due to socioeconomic background, but it is more likely that students prioritize institutions that are geographically closer to their place of residence.

The remaining components (PC5, PC8, PC4, and PC9) are shaped by similar variables, with a strong emphasis on the subjects studied and examined in high school, as well as the educational attainment and occupations of students' parents. This pattern raises important questions about the perpetuation of social and educational inequalities: Does family background reinforce existing social circles, making it more likely for students from certain backgrounds to pursue specific degrees? Are students more inclined to choose a particular field based on the knowledge and exposure gained through their social environment? Is there equal awareness among all students regarding the full spectrum of academic opportunities available to them, or do disparities in information and guidance persist?

5.1.2 Correlation Analysis

Building upon the previous exploration of socioeconomic flows, it was essential to further investigate the relationships between the various variables that shape students' access to university. By analyzing the correlations among these variables, we wanted to identify which associations were strongest, find prevailing trends, and potentially uncover patterns that guide students' pathways into higher education.

The correlation map generated from the dataset revealed the existence of five distinct "hubs" or clusters of variables that are tightly interlinked.

The first hub comprises all variables describing the student's secondary school center, such as the center code, postal code, and indicators of whether the center is public. The correlations within this group approach a value of 1, indicating near-perfect association. This is expected, as these variables are essentially different representations or derivatives of the same underlying entity, the physical secondary school attended by the student. While these variables are informative, their high degree of redundancy means that, if included together in a statistical model, they could introduce multicollinearity and thus distort the analysis.

The second cluster can be found on the right side of the correlation map and brings together the assigned university center and the center where the student ultimately enrolls, with a correlation coefficient of 0.99. This group also includes the code for the vocational training center and its attributes, such as postal code and public/private status. The message here is unequivocal: in the vast majority of cases, students enroll in the same center to which they were initially assigned. Furthermore, the assigned and enrolled centers tend to share both location and legal status with the student's previous institution, indicating that inter-provincial mobility and transitions between public and private institutions are relatively rare.

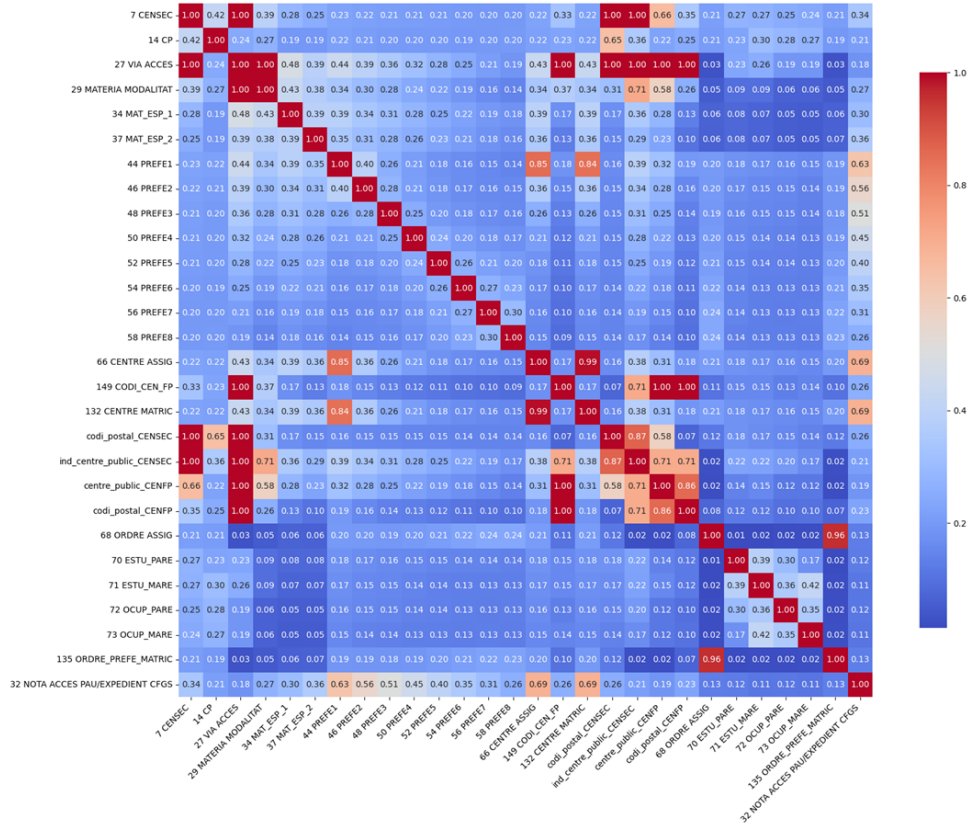


Figure 14: Variable Correlation Results

The third hub consists of the variables related to students' degree preferences). Internal correlations within this group ranged from 0.25 to 0.45 and decreased as the preference positions became more distant (for example, the correlation between the first two preferences is higher than between the first and last one). This pattern suggests a coherent decision-making process: students who select a specific degree as their first choice often choose related fields as their second or third options. However, this coherence diminishes beyond the sixth preference, indicating that lower-ranked choices may be less aligned.

The fourth cluster aggregates the socioeconomic variables of the household, specifically the educational attainment and occupations of the parents. The correlation between the educational levels of both parents is moderate ($r = 0.39$), as is the correlation between education and occupation ($r = 0.30$). The occupations of the father and mother are slightly more closely aligned ($r = 0.42$). These findings reflect a tendency for households to share similar educational and occupational strata, though the correlations are not so high as to suggest homogeneity, since there remains considerable internal diversity.

Finally, the access grade maintains moderate correlations (0.3-0.45) with both the access pathway and the specific subjects studied in the baccalaureate track. This suggests that the pre-university academic itinerary exerts a greater influence on achieving a high access grade than the immediate socioeconomic context. In other words, the route taken through secondary education and the subjects chosen are more predictive of academic performance than family background alone.

Beyond the direct correlations between assigned center and socioeconomic factors, several noteworthy relationships emerged that prompted deeper questions. For instance, the indicator for public vocational and baccalaureate centers shows correlations of 0.58 and 0.71, respectively, with the chosen subject modality. This raises the possibility that students may select subjects or modalities based on their socioeconomic background. While many correlations are not strong enough to confirm this hypothesis definitively, certain values stand out and warrant further investigation.

The access pathway itself has a notable influence on the assigned center ($r = 0.43$), and the public status of the center correlates with the assigned center at 0.38. Furthermore, the indicator for public center

status correlates with first-choice degree preference at 0.38 and 0.32 for baccalaureate and vocational training, respectively. These findings suggest the reinforcement of educational niches and potential segregation, as students from similar backgrounds may cluster in specific pathways or institutions.

Finally, some of the most evident observations reinforce the broader explanation of university access. For example, the correlation between first-choice preference and assigned center is particularly high (0.85), confirming that students generally succeed in entering their preferred programs.

5.2 Mapping Socioeconomic Flows

In order to uncover more precise and quantifiable relationships regarding students' socioeconomic backgrounds, a detailed analysis was conducted linking students' postal codes to their educational trajectories. This process was further enriched by mapping municipal codes to the average family income index (set at 100), thereby allowing for a nuanced understanding of how socioeconomic status may influence access to higher education. Several experiments were carried out to determine whether discernible trends exist in the way students are channeled into university programs: Do certain secondary schools send more students to specific universities? Are there centers that act as major feeders for particular degrees? Is there a tendency for students from certain backgrounds to cluster in specific fields, thereby reinforcing existing patterns and potentially narrowing their academic horizons?

5.2.1 Flows from Secondary Schools to University Degrees: Quantifying Transitions

A central focus of the analysis was to quantify the percentage of students each secondary, baccalaureate, or vocational training center sends to universities, and to identify which university degrees are most frequently chosen by students from particular centers. However, the sheer number of possible combinations between sending and receiving institutions posed a significant challenge, making it difficult to analyze the data at such a granular level. To address this, the data was aggregated in a more interpretable manner, focusing on annual trends and broader patterns rather than isolated cases.

By comparing the rankings of student flows from 2021 to 2024, several consistent axes of continuity emerge, which help explain the movement of students into Catalan universities. Throughout the four years, three main clusters of university studies stand out as the main points of attraction.

- Engineering in Industrial Technologies (including the combined degree and master's program, PARS, from 2022) consistently remains the most popular STEM pathway. Although it briefly ceded prominence to Nursing in 2022, it regained the top position in 2023 and continues to be highly sought after in 2024.
- Nursing (with campuses in Bellvitge, Manresa, Lleida, and Terrassa) experienced a significant surge immediately after the pandemic, accounting for three of the top five connections in 2022 and six appearances in the top 50. While its numbers slightly declined in 2023-24, Nursing remains a natural progression for students from both baccalaureate and, especially, health-related vocational programs.
- Social Work, Social Education, and Early Childhood Education have shown steady growth each year, reaching the top in 2024. Notably, institutions like INS Ronda and INS Vidal i Barraquer now send up to 14% of their students into these fields, reflecting the increased demand for social and educational professionals in the post-COVID labor market.

Rank	Origin Centre	Code	University Degree	Code	Flow	% Global	% Origin
1	INST. RONDA	25006771	Treball social	00061034	16	0.05%	14.55%
2	INSTITUCIO CULTURAL DEL CIC	08032233	Economia	00041014	13	0.04%	6.53%
3	INST. RONDA	25006771	Educació Social	00061013	12	0.04%	10.91%
4	INST. TORRE VICENS	25006732	Infermeria	00061029	12	0.04%	15.79%
5	INST. VIDAL I BARRAQUER	43005704	Treball Social	00071026	12	0.04%	10.71%
6	ESCOLA PIA DE NOSTRA SENYORA	08005242	Disseny “BAU”	00091007	12	0.04%	7.45%
7	INST. CASTELLARNAU	08024893	Infermeria	00021044	12	0.04%	16.44%
8	CAMPUS PROFES-SIONAL UMANRESA	08074999	Educació Infantil (Campus Manresa)	00091038	12	0.04%	46.15%
9	INSTITUCIO CULTURAL DEL CIC	08032233	Eng. en Tecnologies Industrials + M. Eng. Industrial (PARS)	00031115	11	0.03%	5.53%
10	INST. BONANOVA	08035039	Infermeria (Campus Bellvitge)	00011037	11	0.03%	13.10%
11	LA SALLE BONANOVA	08003907	Eng. en Tecnologies Industrials + M. Eng. Industrial (PARS)	00031115	11	0.03%	9.40%
12	INST. CAN VILUMARA	08056109	Treball Social	00011051	11	0.03%	16.42%
13	CULTURAL	08000918	Administració i Direcció d'Empreses	00011033	11	0.03%	11.22%
14	INST. NARCÍS XIFRA I MASMITJÀ	17004499	Infermeria	00081066	11	0.03%	14.47%
15	INST. DE TECNIFICACIÓ	43010888	Ciències de l'Activitat Física i de l'Esport “EUSES”	00071086	10	0.03%	23.81%
16	EASD LLOTJA	08013329	Belles Arts	00011099	10	0.03%	22.73%
17	EAM MASSANA	08038533	Belles Arts	00011099	10	0.03%	21.28%
18	FREDERIC MIS-TRAL-TÈCNIC EULÀLI	08012027	Enginyeria Informàtica	00031059	10	0.03%	5.24%
19	INST. ESCOLA DEL TREBALL-BCN	08013275	Eng. Electrònica Industrial i Automàtica	00031054	10	0.03%	6.54%
20	INST. ESCOLA DEL TREBALL-BCN	08013275	Arquitectura Tècnica i Edificació	00031016	10	0.03%	6.54%
21	LA SALLE GRÀCIA	08007032	Arquitectura Tècnica i Edificació	00031016	10	0.03%	16.13%
22	INST. CASTELLARNAU	08024893	Infermeria EUIT de Terrassa	00021041	10	0.03%	13.70%
23	INST. LA FERRERIA	08044533	Odontologia	00011044	9	0.03%	15.25%
24	FREDERIC MIS-TRAL-TÈCNIC EULÀLI	08012027	Farmàcia	00011024	9	0.03%	4.71%
25	INST. COMTE DE RIUS	43003653	Eng. Electrònica Industrial i Automàtica	00071038	9	0.03%	25.71%
26	BETÀNIA-PATMOS	08004511	Disseny i Innovació “ELISAVA”	00091056	9	0.03%	13.64%
27	INST. RONDA	25006771	Educació Infantil	00061011	9	0.03%	8.18%
28	INST. GAUDÍ	43002570	Treball Social	00071026	9	0.03%	21.43%
29	JESUÏTES SARRIÀ-SANT IGNASI	08004067	Eng. en Tecnologies Industrials + M. Eng. Industrial (PARS)	00031115	9	0.03%	5.26%
30	INST. PERE ALSIUS I TORRENT	17000305	Ciències de l'Activitat Física i de l'Esport “EUSES”	00081082	9	0.03%	14.06%

Table 19: Top 30 Connections: Pre-University Centre - University Degree, year 2024

The data revealed two contrasting patterns among secondary education centers:

- **Highly Specialized Centers.** Some institutions channel the majority of their students into a single academic pathway. For example, CETT sends 68% of its 2021 graduates into Tourism, VITAE Girona directed 58% of its 2022 cohort into Physical Activity Sciences, and Campus Professional UManresa sent 46% of its 2024 graduates into dual Early Childhood Education. These centers act as clear launchpads for specific professional fields.
- **Generalist Centers.** In contrast, academically prestigious and often privately funded centers (such as Jesuïtes Sarrià, Frederic Mistral-Tècnic Eulàlia, and La Salle Bonanova) distribute their student flows across a range of degrees, including Industrial Engineering, Computer Science, and Business Administration. Here, the percentage of students entering any single degree is much lower (around 5% per cohort), suggesting that students from these backgrounds have broader academic options and a more diversified university orientation.

Certain center–university pairings have established themselves as structural corridors, consistently channeling significant numbers of students into specific degrees:

- **INS Torre Vicens to Nursing at UdL:** 18 students in 2022, 17 in 2023, and 12 in 2024, representing up to 25% of the graduating class.
- **INS Ronda to Social Education/Social Work at UdL:** Present in all four years, peaking at 24% in 2023 and remaining the leading pathway in 2024.
- **EAM Massana to Fine Arts at UB:** Always among the top 15 flows, with more than 20% of graduates following this route.
- **Jesuïtes Sarrià, CIC, and La Salle Bonanova to Industrial Engineering at UPC/URL:** These flows, though numerically smaller (9 to 17 students per year), are remarkably stable.

The analysis of flows in relation to the average family income index revealed further insights:

- **High-income, high-cost private schools:** Located in postal codes with above-average income indices, these schools tend to send students into degrees such as Engineering, Architecture, or Business Administration—fields associated with higher earning potential and more demanding entry requirements.
- **Public centers and vocational centers in lower-income areas:** These institutions predominantly channel students into Nursing and Social Work, professions that are in high demand but typically offer moderate salaries and require a strong service orientation.
- **Arts and Hospitality schools:** Institutions like Massana, Llotja, and CETT exhibit an almost one-to-one relationship with their corresponding university degrees, regardless of socioeconomic context. This underscores the strong brand identity and specialized mission of these schools.

Despite the central role of Barcelona in the Catalan university system, geographical proximity remains a decisive factor in student flows. Schools in Girona primarily feed into UdG, those in Lleida into UdL, and centers in Tarragona and Terres de l'Ebre into URV.

The Tecnocampus in Mataró attracts students from local institutes such as Thos i Codina and Santa Anna, particularly in fields like Computer Engineering and Digital Marketing.

5.2.2 Geographic Flows: Tracking Student Movement by Postal Code Over Time

With the enriched dataset providing both the postal codes of students and their educational centers, it was possible to visualize and analyze the geographical flows of students across Catalonia. This approach allowed for a detailed examination of how student mobility patterns evolve over time, the spatial distribution of educational opportunities, and the underlying socioeconomic factors that may influence these flows.

To capture the complexity of these movements, two distinct point maps were generated for each academic year. The first map plots the home postal codes of students, while the second represents the postal codes of the secondary or vocational centers from which they originate. By clustering these points according to their spatial scale, the maps revealed concentrations of students from specific locations, whether these

are their places of residence or the sites of their pre-university education. The resulting clusters provided a visual summary of the density and distribution of student origins and educational backgrounds across the region.

The base map itself was rendered as a choropleth, with color encoding based on the average family income index (set at 100). Given that most areas cluster around the Catalan average, the color scale is segmented more finely for values near the mean, while broader ranges are used for the extremes. This detailed approach ensured that subtle differences in socioeconomic status are visible, particularly in the densely populated middle range, while still highlighting areas of significant affluence or deprivation.

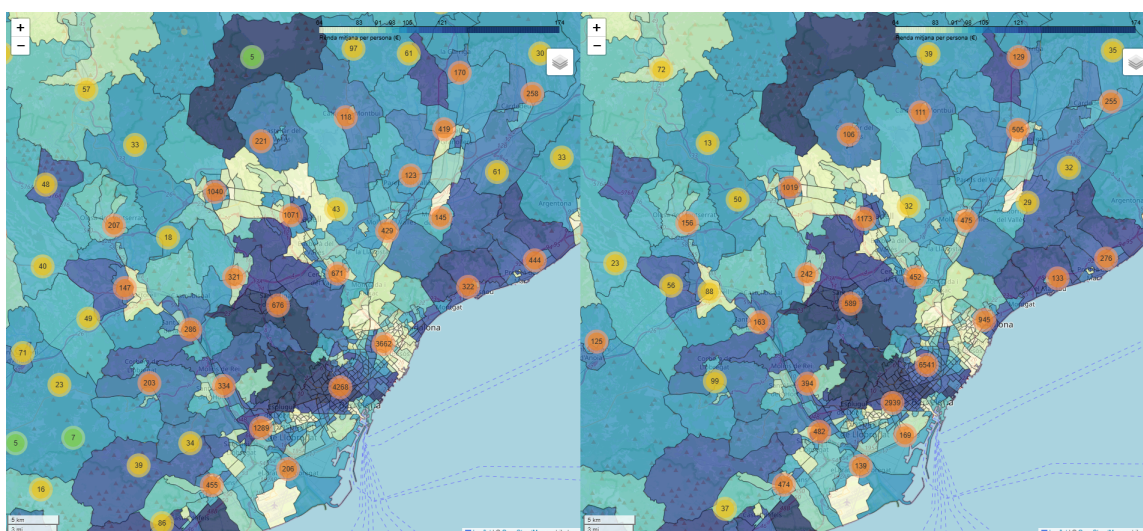


Figure 15: Barcelona Area: On the left, students mapped by home zip code. On the right, students mapped by pre-university center zip code. Year 2024

Analyzing these maps across four consecutive years allowed for the identification of both stable patterns and emerging trends in student mobility. One of the most consistent findings was the strong preference for proximity: students tend to remain close to their home regions when transitioning to university. This tendency supported the hypothesis that geographical closeness remains a key determinant in educational choices, even in a context where major urban centers offer the most prestigious and diverse academic programs.

The maps also made it possible to track changes in the intensity and direction of flows over time. While the overall structure of student movement remains relatively stable, with the largest hubs centered around major metropolitan areas, there are subtle shifts that reflect broader social and economic dynamics. For example, fluctuations in the number of students moving from rural or peripheral areas to urban centers may correspond to changes in transportation infrastructure, housing costs, or the perceived value of certain university degrees.

The general trend is a pronounced concentration of students in and around the major metropolitan areas of Barcelona, Girona, Lleida, and Tarragona, where the leading universities are located. This urban focus is both a reflection of the centralization of academic resources and a potential barrier for students from more remote or economically disadvantaged regions. The cost of living in large cities, including housing, transportation, and daily expenses, can represent a significant financial burden for families, potentially limiting access to high-quality education for those who cannot afford to relocate.

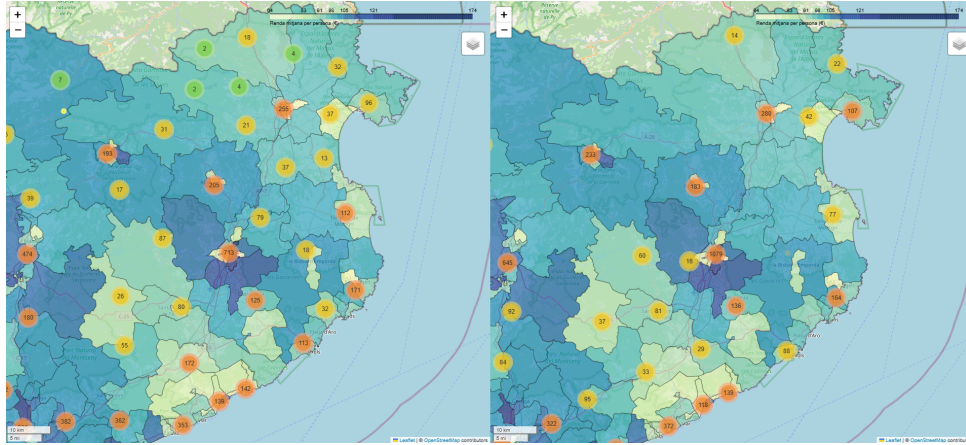


Figure 16: Girona Area: On the left, students mapped by home zip code. On the right, students mapped by pre-university center zip code. Year 2024

This centralization of opportunity has broader implications for social equity and regional development. While urban centers benefit from a continuous influx of talent and innovation, rural and less affluent areas may experience a drain of their most ambitious students, exacerbating existing disparities. The maps highlight not only where students are coming from, but also the socioeconomic contexts they leave behind, raising important questions about the distribution of educational opportunity and the long-term effects on regional cohesion.

Despite the dominance of Barcelona as the principal academic hub, the data also reveal a network of secondary centers distributed throughout Catalonia. These regional institutions play a crucial role in providing access to higher education for students who may be unable or unwilling to move to the capital. The flows to these centers, while smaller in absolute numbers, are vital for maintaining educational diversity and supporting local economies.

The similarity in the number of students moving between regions each year suggests a degree of stability in the system, but the underlying motivations and constraints may vary. For some, the decision to stay close to home is driven by financial necessity; for others, it reflects a desire to remain within familiar social and cultural environments. The maps thus serve as a starting point for more in-depth analysis of the factors shaping student choices, including family background, academic aspirations, and perceptions of opportunity.

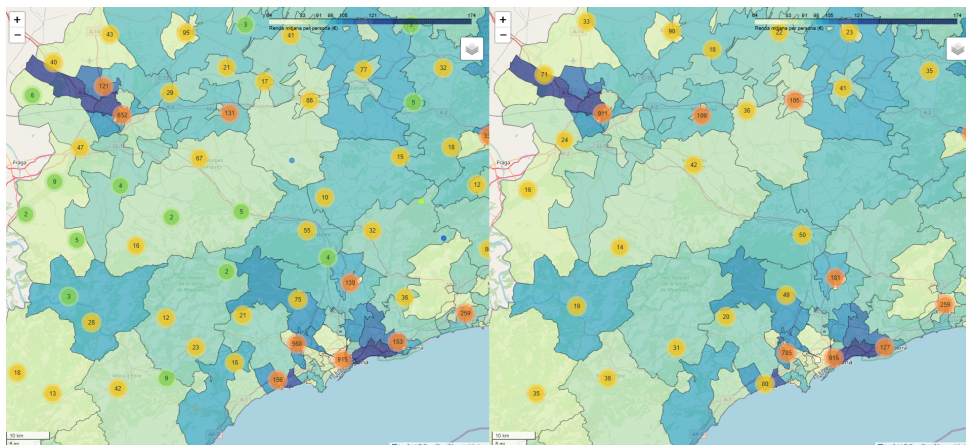


Figure 17: South of Catalonia: On the left, students mapped by home zip code. On the right, students mapped by pre-university center zip code. Year 2024

5.2.3 Linking Family Income Index to Degrees: Identifying High-Income Degrees, Annual Rankings and Social Niches

The initial analysis of student flows and socioeconomic backgrounds naturally led to a more focused investigation: mapping the average family income index (index 100) by postal code and linking this information to specific university degrees. The objective was to determine whether certain academic programs consistently attract students from more privileged backgrounds, thereby revealing potential social and economic niches within the higher education system. This mapping is not merely descriptive. It seeks to identify which degrees have the highest average family income among their student cohorts each year, and to assess whether these patterns are stable or subject to change over time. By calculating the average IRF for each degree from 2021 to 2024, several measurable regularities emerged, based solely on the income index, cohort size, and relative ranking. The data revealed a clear and persistent stratification within the university system.

First, the absolute leader in terms of family income was unchanging: the integrated degree and master's program in Industrial Technologies and Economic Analysis consistently occupied the top position. Its average IRF remains exceptionally high (ranging from 121 to 125 over the four years) while the group size is remarkably stable, fluctuating only slightly between 44 and 47 students. This consistency highlights a sustained concentration of high-income students in this program, making it the most exclusive in socioeconomic terms.

The second most notable case is the standalone degree in Industrial Technologies Engineering (PARS), which, even without the Economic Analysis track, combines the highest volume of students with a high average income. With annual groups ranging from 534 to 614 students and an IRF that rises from 112 to 115, this degree is unique in appearing simultaneously in the top ten for both income and student volume. It thus serves as the central axis where magnitude and privilege intersect.

A closer look at the top-50 degrees by income each year revealed a stable core: 31 programs remained in this elite group throughout the four-year period. Of these, 18 are various engineering degrees, 7 were in business, administration, or marketing, and 6 were in design or the arts. Together, these three fields accounted for over 60% of all high-income entries annually. The dominance of engineering was particularly striking, with 19 to 21 entries each year, followed by business or marketing (10-12) and design and arts (7-9). Other areas, such as health sciences, humanities, and pure sciences, collectively represented less than a third of the high-income positions, and this distribution remains largely unchanged over time.

Design degrees at ELISAVA (and, to a lesser extent, at BAU and Eina) maintained consistently high IRF values, typically around 115-117, even as their cohort sizes fluctuated slightly. For example, the Design and Innovation program at ELISAVA enrolls between 144 and 166 students annually. This stability suggests a sustained, though moderate, concentration of high-income students in these programs.

Rank	Degree	Avg. Income	Sample (n)
1	Tecnologies Industrials amb Anàlisi Econòmica, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial amb Anàlisi Econòmica) (docència en Anglès)	122	44
2	Enginyeria en Organització Industrial “EU Salesiana de Sarrià”	119	33
3	Enginyeria de Disseny Industrial “ELISAVA”	117	66
4	Enginyeria en Tecnologies Industrials, Grau en + Màster Universitari en Enginyeria Industrial (PARS: Enginyer/a Industrial)	115	549
5	Disseny i Innovació “ELISAVA”	115	151
6	Filosofia, Política i Economia	115	27
7	Enginyeria de Tecnologies i Serveis de Telecomunicació, Grau en + Màster Universitari en Enginyeria de Telecomunicació (PARS: Enginyer/a de Telecomunicació)	114	38
8	Administració i Direcció d’Empreses / Matemàtiques (simultaneity)	114	21
9	Estudis Àrabs i Hebreus	114	4
10	Direcció i Gestió Gastronòmica “Barcelona Culinary Hub”	114	4
11	Farmàcia / Nutrició Humana i Dietètica (simultaneity)	113	13
12	Administració i Direcció d’Empreses / Sociologia (simultaneity)	113	21
13	Enginyeria Electrònica Industrial i Automàtica “EU Salesiana de Sarrià”	113	20
14	Ciència i Enginyeria de Dades	113	43
15	Enginyeria Electrònica Industrial i Automàtica / Enginyeria Mecànica “EU Salesiana de Sarrià” (simultaneity)	112	12
16	Administració d’Empreses i Gestió de la Innovació “Tecnocampus” (docència en Anglès)	112	29
17	Màrqueting “ESIC”	112	66
18	Enginyeria d’Automoció “EU Salesiana de Sarrià”	112	44
19	Ciència, Tecnologia i Humanitats	112	18
20	Administració i Direcció d’Empreses – Economia / Dret (simultaneity group)	112	83

Table 20: Top 20 Degrees by Highest Average Income, year 2024

Double degrees and integrated programs exhibit higher average family incomes than their single-degree counterparts. For instance, the joint degree in Business Administration and Mathematics records an IRF of 117 in 2021, dips to 111 in 2022, and then rises again to 114-115 in subsequent years, with consistently small cohorts of fewer than 25 students. Similar patterns are observed in other double degrees, such as Law and International Relations or Pharmacy and Nutrition, confirming that these combinations tend to attract students from more affluent backgrounds, albeit in limited numbers.

In contrast, degrees that dominate in terms of student volume, such as Nursing, Early Childhood Education, and Social Work, were entirely absent from the top-50 income rankings. Their average family incomes fall below the minimum threshold of the first positions, indicating that these fields could be attracting students from less privileged backgrounds.

Annual fluctuations in the composition of the top-50 are modest, with only 5 to 7 new entries each year, and the income threshold remains virtually unchanged (IRF 108-109). For example, the degree in Artificial Intelligence enters the ranking in 2023 with an IRF of 109 and a cohort of 43 students, maintaining its position in 2024 without displacing any of the traditional high-income programs.

Finally, the GCED degree exemplifies this pattern, appearing among the top positions year after year, with the number of students enrolled varying but its high-income status remaining constant (placing 7th in the ranking in 2021, 26th in 2022, 11th in 2023 and 14th in 2024).

Various intertwined mechanisms could help explain why the programme systematically draws high-income students and how it differs from the usual degrees at the very top (Industrial Technologies/PARS, Business and Mathematics, ELISAVA design).

For instance, GCED requires one of the highest cut-off scores within its faculty and explicitly recommends the Technology or Science baccalaureate track plus an A-level in calculus with competitive grades. Those prerequisites favour pupils from well-resourced schools and families that can finance private tutoring or international curricula, exactly the group that already populate Industrial Technologies and the Business and Math cluster.

On the other hand, regarding the students's closest social environments, parents and counsellors perceive data science as the "21st-century medicine": global work demand and high salaries within five years. The same salary tendency is observed in Integrated Industrial Engineering, but GCED couples it with the glamour of AI, boosting its appeal in affluent, career-oriented households.

5.3 Interpreting Patterns: Dynamic, Static, and Hermetic Degree Profiles

To deepen the analysis of social and territorial diversity within university programmes, we moved beyond aggregate income indices and examined the specific origins of students as indicated by their postal codes. Our central question was whether certain degrees consistently attracted students from the same backgrounds or whether genuine diversity existed in the origins of their groups. We addressed this by calculating, for each degree, the frequency with which every postal code appeared among its students, thereby constructing a postal code histogram for each programme. After normalising these counts, we obtained a distribution that reflected the territorial diversity of every degree. The key metric of interest became the variance (or, more generally, the variability) of these distributions: higher variance pointed to a broader range of backgrounds, whereas lower variance suggested a more homogeneous intake.

Once we had established the postal code distributions for each degree, we computed the variance (or another measure of variability) per programme. We then ranked degrees from the lowest to the highest variance and identified the top 50. This ranking yielded a clear, quantitative picture of which programmes were most territorially diverse and which were most homogeneous in terms of student origins.

To refine the analysis, we calculated two complementary metrics: one based on the raw counts of postal codes and another using the normalised histogram distributions. The first metric, Intersection over Union (IoU), measured the stability of the set of postal codes present in a degree from year to year. A high IoU indicated that nearly the same set of postal codes appeared each year, suggesting a stable and possibly exclusive recruitment pattern. The second metric, Jensen–Shannon divergence (JSD), captured changes in the relative weights of each postal code within the distribution. A low JSD meant that not only were the same postal codes present, but their proportions remained similar over time. A high JSD showed that,

although the set of postal codes might have stayed stable, their relative representation shifted markedly.

- Low IoU, Low JSD: Degrees in this quadrant proved the most territorially dynamic. They experienced significant turnover in the postal codes represented and in the proportions of students from each area, reflecting high openness and social mobility.
- High IoU, Low JSD: These programmes were diverse in the number of postal codes represented, yet that diversity remained static. The same areas appeared each year and in nearly the same proportions, hinting at a "frozen" diversity.
- High IoU, High JSD: Such degrees functioned as nearly hermetic circles: the set of postal codes was almost immutable, but the internal share of students from each code fluctuated, indicating shifts within a closed group rather than genuine openness.

Business Administration (ADE), Primary Education and Early Childhood Education, all exhibited high IoU values (≥ 0.41). We observed that almost the same postal codes were represented every year. In ADE, for instance, a low JSD ($\text{JSD} = 0.223$) showed that the proportions of students from each area hardly changed, pointing to a diverse yet remarkably stable intake.

Nursing at Bellvitge, Psychology, Social Work represent vocational degrees with centralised recruitment, showed high IoU (0.39-0.41) but moderate to high JSD (0.43-0.48). The same neighbourhoods always appeared, yet their relative representation varied annually, possibly owing to shifts in admission criteria, capacity, or scholarship policies.

Industrial Engineering (PARS) exemplified a closed, rotating niche. With an IoU of 0.407, it maintained a fixed core of high-income postal codes (Sarrià, Sant Gervasi, Valldoreix). A JSD above 0.46 indicated that the internal distribution among these affluent areas shifted, yet new, less privileged areas were not incorporated.

Territorial clusters of engineering programmes in Terrassa, Sabadell, and Igualada posted IoU values around 0.389 and $\text{JSD} = 0.361$. Classic industrial postal codes remained constant, but their relative importance changed as different engineering specialisations rose or fell in labour-market appeal.

Degrees such as Telecommunications Systems Engineering or general Nursing displayed high IoU and high JSD, experiencing abrupt expansions and contractions in territorial reach. New postal codes appeared one year and disappeared the next, driving large entropy variations (ΔH up to 0.789) and revealing sensitivity to external factors such as policy changes or demand shifts.

Labour Relations and related programmes showed limited but highly regular variety: $\text{IoU} \approx 0.37$ with low JSD and ΔH . These degrees consistently drew from a small set of urban, middle-class postal codes in nearly fixed proportions, indicating a stable niche recruitment pattern.

We found that programme size did not necessarily correlate with territorial stability. ADE and Primary Education, each enrolling thousands of students, exhibited minimal variability, whereas Nursing, of similar cohort size, displayed higher JSD. The crucial factor was the dispersion of each programme's offerings (many campuses versus a single campus) and the nature of the pre-university recruitment channel. ADE, for example, appeared in multiple variants (public, EUNCET, Tecnocampus, Manresa), each reproducing a similar IoU-JSD profile and local postal code pattern. Campus location added a geographic nuance but did not fundamentally alter the underlying logic.

In this scenario, GCED sits in the lower-middle of the overall stability ranking (240th place). Its average Jensen-Shannon divergence of 0.736 bits signals that the mix of students' home postcodes shifts markedly from one group to the next. Although the absolute change in postcode-entropy ($|\Delta H| \approx 0.25$ bits) shows only modest swings in overall diversity, the year-pair breakdown reveals that each successive intake is drawn from a distinctly different geographic profile: 0.702 from 2021 to 2022, 0.797 from 2022 to 2023, and 0.709 from 2023 to 2024. The jump from 2022 to 2023 is the most pronounced, accounting for the largest share of the "instability" and suggesting that a factor such as a change in admissions criteria, program marketing, or competing offerings altered admissions that year.

6 Recommender Systems

Building on our multifaceted analysis, the recommender-system module aimed to translate observed patterns of student preferences into personalized, ranked study suggestions. The recommender is designed to mirror the eight-slot preference vectors that applicants themselves fill out, but to do so automatically, drawing on historical assignment behavior and the rich set of student and program attributes we have assembled. Rather than prescribing a one-size-fits-all ordering, we wanted the system to construct a tailored sequence of program recommendations for each user, thereby reflecting collective trends and adapting to individual profiles.

In its most elemental design, the system exploits co-occurrence signal, essentially the frequency with which two programs appear together in the same student’s preference list. By tabulating these joint appearances across the entire dataset, we derive a popularity-weighted network of program affinities. To generate recommendations, we simply rank a target student’s unselected programs by their co-occurrence strength with the student’s indicated partial preferences, thereby filling empty slots with the most “peer-tested” choices. This “popularity” model provides a transparent baseline: it leverages the knowledge of past students, requires minimal parameter tuning, and offers immediate insights into which programs are widely considered complementary in actual application behavior.

6.1 First Versions

In our initial approach to recommendation modeling, we adopted two complementary paths that reflect progressively deeper levels of pattern extraction from the same underlying preference data. The first, which we refer to as the co-occurrence baseline, rests on a straightforward, but surprisingly powerful, intuition: programs that frequently appear together in students’ ranked lists are, in some sense, “affine” to one another. To operationalize this, we aggregate over the historical preference vectors of all students and tally every pair of programs that co-occur within the same individual’s choices. The result is a symmetric matrix of co-occurrence counts that directly captures the affinity strength between any two offerings. When it comes to generating a suggestion list for a new student—who might only have partially filled out their own preference vector, we simply look up the programs they have already selected, fetch all their co-occurrence partners from the matrix and rank these candidates by their total weight. Empty slots are filled in descending order of these aggregated affinities, while ensuring no duplicates. This method is transparent, easy to compute, and immediately yields “crowd-endorsed” recommendations without any parameter tuning.

Although the co-occurrence method has the virtues of simplicity and interpretability, it treats each program pair in isolation and does not account for higher-order patterns in the data. To capture more latent structure in the student–program interaction landscape, we next turned to matrix-factorization via Singular Value Decomposition (SVD). In this framework, we view the entire set of historical preferences as one big interaction matrix, with rows representing students and columns representing program slots. Each cell encodes whether or how strongly a student expressed interest in a given program. By performing a truncated SVD on this matrix, we decompose it into a small number of latent factors that jointly explain most of the observed co-selection patterns. In effect, both users and items are embedded into a shared low-dimensional space where geometric proximity reflects underlying thematic or behavioral affinities.

Once the SVD embeddings are learned, generating recommendations becomes a matter of projecting a student’s partial profile into the latent space by summing or averaging the factor vectors corresponding to their chosen programs and then retrieving the nearest-neighbor programs according to cosine similarity in that space. Unlike the raw co-occurrence counts which only know about directly observed pairs, the latent embeddings can infer indirect relationships: two programs that have never co-occurred may nonetheless lie close together if they co-occur with the same third set of programs. This ability to generalize beyond the training examples makes the SVD approach more robust to sparsity and more sensitive to the deeper structure of student choice behavior.

Comparing the two methods, we find that the co-occurrence baseline excels at surfacing the most popular and intuitive pairings, useful for a quick “most-likely” suggestion, but tends to over-recommend the same handful of dominant programs. In contrast, the SVD recommender offers a broader variety of options by leveraging latent dimensions that capture niche or emergent affinities, albeit at the cost of

added complexity and reduced transparency. As such, these first-version models form a natural duo: the co-occurrence approach provides an immediately deployable result based on popularity, while the SVD factorization lays the groundwork for more sophisticated, personalized rankings that can incorporate side information or be further refined with regularization and feedback loops.

Some initial results we obtained:

```
query = np.array([61069, 21014, 11033, 0, 0, 0, 0, 0])
```

Student	Code	University	Name
1	61069	UdL	Administració i Direcció d'Empreses
2	21014	UAB	Administració i Direcció d'Empreses
3	11033	UB	Administració i Direcció d'Empreses
4	11103	UB	Economia
5	41008	UPF	Administració i Direcció d'Empreses
6	21070	UAB	Economia
7	41014	UPF	Economia
8	11012	UB	Dret

Table 21: Popular-based Recommendations

Student	Code	University	Name
1	61069	UdL	Administració i Direcció d'Empreses
2	21014	UAB	Administració i Direcció d'Empreses
3	11033	UB	Administració i Direcció d'Empreses
4	11103	UB	Economia
5	11012	UB	Dret
6	21084	UAB	Comptabilitat i Finances
7	21099	UAB	Administració i Direcció d'Empreses (docència semipresencial)
8	41009	UPF	Ciències Empresarials – Management

Table 22: Popular-based (Unique Names)

We then went on to test our final SVD recommender:

Student	Code	University	Name
1	71016	URV	Infermeria
2	71028	URV	Infermeria
3	71032	URV	Infermeria
4	11006	UB	Ciències Biomèdiques
5	21041	UAB	Infermeria (EU Terrassa)
6	21010	UAB	Ciències Biomèdiques
7	61029	UdL	Infermeria
8	21042	UAB	Infermeria “Gimbernat”

Table 23: SVD Student A Recommendations

Student	Code	University	Name
1	11097	UB	Comunicació Audiovisual
2	21085	UAB	Comunicació Audiovisual
3	41008	UPF	Administració i Direcció d'Empreses
4	41002	UPF	Comunicació Audiovisual
5	41004	UPF	Publicitat i Relacions Públiques
6	21097	UAB	Publicitat i Relacions Públiques
7	21096	UAB	Periodisme
8	11093	UB	Publicitat, Relacions Públiques i Màrqueting

Table 24: SVD Student B Recommendations

6.2 Simple Machine Learning Models

In this phase, we explored the potential of straightforward supervised learning models to predict a student's likely assignment based on their partial preference profile and academic indicators. The core idea is to treat the assignment task as a multiclass classification problem, where each program represents one of hundreds of possible target classes. By training models such as random forests and multinomial logistic regression on historical data, we aim to capture the implicit decision rules that link a student's expressed interests, performance metrics, and modest background attributes to the eventual outcome of their application process.

Our modeling logic rests on two complementary principles. First, we prioritize features that reflect the student's own aspirations and readiness: their ranked preferences (as a fixed-length vector, padded where necessary), specialized subject choices, and standardized access scores. These signals encode both the student's thematic orientation and their academic qualification, which are the most direct drivers of assignment. Second, we include a small set of contextual variables, notably the student's initial admission pathway and the type of institution they ultimately joined, to provide minimal structural context without letting geography or socioeconomic status dominate. By consciously limiting the model's inputs to these categories, we ensure that recommendations are driven primarily by alignment between student aptitude, interest, and program profiles.

The random forest classifier, with its ensemble of decision trees, excels at capturing nonlinear interactions among these features, such as how a certain combination of specialty-subject grades and a second-choice preference might tilt the likelihood toward one cluster of programs over another. In training, we balance class frequencies to mitigate the dominance of highly popular programs, ensuring the model remains sensitive to less common but still meaningful applicant trajectories.

In parallel, the multinomial logistic regression provides a transparent, parametric perspective: here, each feature contributes additively to the log-odds of a student landing in any specific program. The model's regularized coefficients help prevent overfitting in the face of hundreds of target categories and thousands of sparse feature encodings. Moreover, the logistic framework naturally yields probability scores across all programs, which we can rank to produce an ordered recommendation list of eight study options.

But how do we make sure we are not reinforcing biased patterns? To keep the recommender from simply reproducing the structural advantages present in the historical data, we follow a series of "fairness rules".

- Any variable that is a direct or obvious proxy for household wealth, parental education, or local schooling resources (postal code, sending-school identifier, parental occupation, grant status) is excluded from both training and inference. What remains are the student's own choices (ranked list and specialised subjects) and performance signals (access mark, specific-subject grades). In other words, every learner starts from a "fresh slate" in the eyes of the model, regardless of social or family background.
- Because biased inputs yield biased outputs, we judge every candidate model not only by top-k accuracy but also by opportunity metrics: degree-set diversity, novelty and coverage across disciplines while maintaining similarity and closeness. A model that is one or two points less "accurate"

yet offers a wider palette of credible programs is preferred over a hyper-accurate but narrow, self-confirming predictor.

Evaluation on held-out students demonstrates that both models achieve strong alignment with actual assignments, with the random forest typically offering marginally higher top-choice accuracy and the logistic model producing smoother probability distributions that capture second and third-choice behavior more faithfully. Qualitative inspection of the recommendation lists confirms that the models respect individual student profiles, suggesting advanced engineering programs for candidates with high specialty scores and pertinent electives, while proposing life-science degrees for those whose grades and early preferences tilt in that direction.

Together, these simple yet robust classifiers establish a baseline that leverages readily available student data and can be readily extended with more elaborate features or blended with the co-occurrence and latent-factor approaches previously described. They demonstrate that even relatively lightweight machine learning tools can produce personalized, ranked study option recommendations by learning directly from historical assignment patterns.

Some initial results we obtained:

Student	Code	University	Name
0	71006	URV	Biotecnologia
	61005	UdL	Biotecnologia
	81006	UdG	Biologia
	11024	UB	Farmàcia
	91054	UVic-UCC	Medicina
	61030	UdL	Medicina
	71054	URV	Bioquímica i Biologia Molecular / Biotecnologia
	71024	URV	Química
1	71002	URV	Administració i Direcció d'Empreses
	71014	URV	Finances i Comptabilitat
	71004	URV	Administració i Direcció d'Empreses
	71008	URV	Dret
	71068	URV	Gestió en Turisme i Hoteleria
	71016	URV	Infermeria
	61003	UdL	Administració i Direcció d'Empreses
	11033	UB	Administració i Direcció d'Empreses
2	81057	UdG	Arquitectura, Estudis d'
	31078	UPC	Arquitectura, Estudis d'
	81065	UdG	Arquitectura Tècnica i Edificació (semipresencial)
	81020	UdG	Arquitectura Tècnica i Edificació
	31016	UPC	Arquitectura Tècnica i Edificació
	11047	UB	Psicologia
	11002	UB	Biologia
	21071	UAB	Enginyeria Informàtica
3	91050	UVic-UCC	Criminologia "ESERP"
	91057	UVic-UCC	Enginyeria de Disseny Industrial "ELISAVA"
	10120	UB	Investigació Privada (Títol propi, no oficial)
	91915	UPC/UPF	Tecnologies Industrials amb Anàlisi Econòmica + Màster
	91914	UB/UPC	Economia / Estadística (simultaneïtat)
	91913	UAB/UB	Filologia i Cultura Franceses
	91912	UPF/UB/UPC/UAB	Bioinformàtica "ESCI"
	91909	UdG/UB	Fisioteràpia (només no-residents)
4	71064	URV	Enginyeria Biomèdica
	11002	UB	Biologia
	31059	UPC	Enginyeria Informàtica
	21044	UAB	Infermeria
	61074	UdL	Enginyeria Informàtica
	71005	URV	Bioquímica i Biologia Molecular
	71006	URV	Biotecnologia
	11024	UB	Farmàcia

Table 25: Random Forest Recommendations by Student

Student	Code	University	Name
0	71006	URV	Biotecnologia
	11075	UB	Farmàcia / Nutrició Humana i Dietètica
	71054	URV	Bioquímica i Biologia Molecular / Biotecnologia
	11004	UB	Biotecnologia
	21035	UAB	Genètica
	81066	UdG	Infermeria
	31067	UPC	Enginyeria Física
	41033	UPF	Enginyeria Biomèdica
1	71002	URV	Administració i Direcció d'Empreses
	71014	URV	Finances i Comptabilitat
	71082	URV	Relacions Laborals i Recursos Humans
	71026	URV	Treball Social
	61053	UdL	Administració i Direcció d'Empreses / Turisme
	71068	URV	Gestió en Turisme i Hoteleria
	61003	UdL	Administració i Direcció d'Empreses
	71004	URV	Administració i Direcció d'Empreses
2	81057	UdG	Arquitectura, Estudis d'
	31103	UPC	Enginyeria Civil
	81020	UdG	Arquitectura Tècnica i Edificació
	31016	UPC	Arquitectura Tècnica i Edificació
	31113	UPC	Enginyeria Civil + Màster en Camins, Canals i Ports
	31078	UPC	Arquitectura, Estudis d'
	31087	UPC	Enginyeria en Geoinformació i Geomàtica
	31027	UPC	Enginyeria Mecànica
3	91050	UVic-UCC	Criminologia "ESERP"
	71062	URV	Màrqueting "ESIC"
	31095	UPC	Ciències i Tecnologies de l'Esport
	31002	UPC	Administració i Direcció d'Empreses "EAE"
	31090	UPC	Màrqueting i Comunicació Digital "EAE"
	81014	UdG	Gestió Hotelera i Turística
	61033	UdL	Relacions Laborals i Recursos Humans
	31082	UPC	Màrqueting i Comunicació Digital "EUNCET"
4	71064	URV	Enginyeria Biomèdica
	71006	URV	Biotecnologia
	11030	UB	Física
	11006	UB	Ciències Biomèdiques
	61040	UdL	Ciència i Producció Animal / Veterinària
	21100	UAB	Física / Química
	21083	UAB	Veterinària
	21035	UAB	Genètica

Table 26: Logistic Regression Recommendations by Student

7 Conclusions and Future Work

7.1 Future Work: Complex Model Proposals

The pipeline presented in this thesis favours models that balance predictive power with engineering simplicity. However, as recommender system research accelerates, markedly more expressive deep-learning architectures have begun to dominate public leaderboards, suggesting a clear path for future work to continue this initial investigation. This final section sketches two directions that could provide a new perspective: encoder-decoder sequence models and graph neural networks. It is important to highlight the scientific questions they open, the resources they demand, and the practical hurdles that must be cleared before deployment.

7.1.1 Encoder-Decoder Architectures

Encoder-decoder (seq2seq) frameworks treat a user's interaction history as a language in which items are "tokens" and sequences encode behavioural context. Modern implementations leverage Transformer blocks with multi-head self-attention, enabling the encoder to build rich, position-aware embeddings of arbitrarily long click streams, while the decoder autoregressively generates a slate of next items. Such models promise three concrete gains:

1. **Long-Range Dependencies:** Self-attention can relate events that occur dozens or hundreds of steps apart, capturing seasonal interests and slow-evolving patterns that shallow models miss.
2. **Generative Candidate Creation:** Instead of re-ranking a fixed catalogue slice, the decoder can propose entirely new item combinations.
3. **Self-Supervised Pre-Training:** Masked-item or next-item prediction objectives allow unsupervised pre-training on billions of events, after which the model can be fine-tuned with modest labelled data.

The key challenges are computational: attention complexity scales quadratically with sequence length, and beam search inference can inflate latency. Promising mitigations include sparse or linearised attention kernels, knowledge distillation into lighter student networks, and hybrid cascades in which a cheap model filters candidates before the encoder-decoder refines the final list.

7.1.2 Graph Neural Networks

Graph Neural Networks (GNNs) extend the message-passing paradigm to the bipartite (or heterogeneous) graph formed by users, items, and side entities such as tags or brands. By iteratively aggregating information from a node's L -hop neighbourhood, GNNs learn representations that encode high-order collaborative signals ("users similar to you but two hops away liked item i ") as well as attribute similarities and knowledge-graph semantics.

Incorporating GNNs raises several research avenues:

1. **Dynamic Graph Modelling:** Real-world interaction graphs evolve continuously. Temporal GNN variants (e.g. TGAT, DyGNN) could capture novelty effects and drifting preferences.
2. **Scalability and Sampling:** Large-scale graphs exceed GPU memory. Neighbourhood-sampling strategies (GraphSAGE, PinSAGE) or approximate personalised PageRank propagation can keep training tractable without severe accuracy loss.
3. **Explainability and Fairness:** Attention weights or gradient-based saliency scores offer local explanations of why a recommendation surfaced, while adversarial regularisers could mitigate popularity bias and ensure equitable exposure across item groups.

Deploying GNNs would require an end-to-end pipeline for efficient mini-batch graph sampling and feature mapping, as well as a rigorous testing protocol to measure real-world gains against their higher serving cost.

Both encoder-decoder and GNN paradigms demand deeper networks, larger parameter counts, and non-trivial infrastructure changes, yet they align with the current state of the art in recommendation research.

7.2 Conclusions

Through its multifaceted approach, this thesis not only reveals the structural and social variables at play in university admissions but also proposes practical tools and methodologies to foster greater diversity and opportunity within the academic landscape. The findings highlight the need for targeted policy interventions, such as improved information dissemination, guidance, and support for underrepresented groups, in order to democratize access to university programmes.

The analysis confirms that access to higher education in Catalonia is shaped by a complex interplay of institutional, economic, and social factors. Socioeconomic disparities are closely linked to both the opportunities available to students and the diversity of student profiles within each degree. High-income students are systematically overrepresented in certain degrees (especially in engineering, business, and design) while other fields, such as nursing and social work, attract students from less privileged backgrounds.

The mapping of student flows from secondary schools to university degrees reveals persistent patterns of both specialization and generalization. Some secondary centers act as highly specialized feeders into specific university programmes. Geographical proximity remains a decisive factor in university choice, with most students preferring to study close to home, reinforcing the centralization of opportunity in major urban areas.

On another note, the construction and analysis of degree knowledge graphs, based on co-occurrence patterns in student preferences, have illuminated the evolving structure of academic disciplines. The network analysis identified stable clusters corresponding to disciplinary families (e.g., STEM, health sciences, business, humanities), as well as the emergence of new interdisciplinary hubs (e.g., Smart Cities, Bioinformatics).

This could be the starting point of further analysis to identify patterns over the next years, especially in light of big systemic and educational changes regarding the importance AI will play in academia.

Finally, we can affirm our observation and hypothesis are on the right track. Beyond formal criteria such as grades and entrance exams, "hidden curricula of opportunity" does exist and access to information, guidance, and social networks plays a crucial role in shaping academic aspirations and outcomes. Not all students are equally aware of the full spectrum of academic opportunities, especially in emerging fields, leading to the reproduction of professional elites and the concentration of talent in specific sectors.

As we reflect on the findings of this thesis, several questions remain. How can we make sure that pathways to higher education in Catalonia can one day be open to all, regardless of socioeconomic background or access to information? What further steps can institutions take to break down the invisible barriers that perpetuate inequality and limit diversity within our universities? What action should we take to shift entrenched patterns? Is a broader cultural and policy transformation required?

References

- [1] Ley orgánica 2/2006, de 3 de mayo, de educación. BOE núm. 106, de 4 de mayo de 2006.
- [2] Ley orgánica 3/2020, de 29 de diciembre, por la que se modifica la ley orgánica 2/2006, de 3 de mayo, de educación. BOE núm. 340, de 30 de diciembre de 2020.
- [3] Llei 12/2009, del 10 de juliol, d'educació. DOGC núm. 5422, de 16 de juliol de 2009.
- [4] Sergey Brin and Lawrence Page. The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1-7):107–117, 1998.
- [5] Aparna Dhinakaran. How to understand and use the jensen-shannon divergence. How to Understand and Use the Jensen-Shannon Divergence, 2023.
- [6] Distilled AI. Recommendation systems (recsys). Recommendation Systems (RecSys), 2024.
- [7] David Easley and Jon Kleinberg. *Networks, Crowds, and Markets: Reasoning About a Highly Connected World*. Cambridge University Press, 2010.
- [8] Educaweb. Estructura i nivells del sistema educatiu. Estructura i nivells del sistema educatiu, 2024.
- [9] Geeks for Geeks. Hierarchical clustering in machine learning. Hierarchical Clustering in Machine Learning, 2025.
- [10] Generalitat de Catalunya. Sistema educatiu a catalunya, curs 2025-2026. Sistema educatiu a Catalunya, curs 2025-2026, 2024.
- [11] IBM. What is random forest? What is random forest?, 2025.
- [12] IDESCAT. Indicadors socioeconòmics estandarditzats. Indicadors socioeconòmics estandarditzats, 2021.
- [13] IDESCAT. Ensenyament universitari. alumnes matriculats. per sexe. curs 2021/2022. Ensenyament universitari. Alumnes matriculats. Per sexe. Curs 2021/2022, 2022.
- [14] IDESCAT. Ensenyament universitari. alumnes titulats. per sexe. curs 2021/2022. Ensenyament universitari. Alumnes titulats. Per sexe. Curs 2021/2022, 2022.
- [15] IDESCAT. Nivell de formació assolit per la població de 15 anys o més. per sexe i nivells de formació. Nivell de formació assolit per la població de 15 anys o més. Per sexe i nivells de formació, 2022.
- [16] IDESCAT. Índex socioeconòmic territorial. Institut d'Estadística de Catalunya. Índex socioeconòmic territorial, 2024.
- [17] Scikit learn. Randomforestclassifier. RandomForestClassifier, 2007-2025.
- [18] Medium. Bart4rec: Sequence-to-sequence generative model for sequential recommender system. BART4Rec: Sequence-to-Sequence Generative Model for Sequential Recommender System, 2024.
- [19] M.E.J. Newman. Modularity and community structure in networks. *roc. Natl. Acad. Sci. U.S.A.* 103 (23) 8577-8582, 2006.
- [20] Observatorio del Sistema Universitario. Notas de acceso a la universidad: ¿son equitativas? Notas de acceso a la universidad: ¿son equitativas?, 2023.
- [21] Shashank Rajput, Nikhil Mehta, Anima Singh, Raghunandan Keshavan, Trung Vu, Lukasz Heldt, Lichan Hong, Yi Tay, Vinh Q. Tran, Jonah Samost, Maciej Kula, Ed H. Chi, and Maheswaran Sathiamoorthy. Recommender systems with generative retrieval. *37th Conference on Neural Information Processing Systems*, 2023.
- [22] Hamid Rezatofighi, Nathan Tsoi, JunYoung Gwak, Amir Sadeghian, Ian Reid, and Silvio Savarese. Generalized intersection over union. June 2019.
- [23] Britta Ruhnau. Eigenvector-centrality — a node-centrality? *Symposium on Operations Research (SOR) in Magdeburg*, 1999.

- [24] Caroline Scharf. From theory to practice: Applying graph theory network analysis. From Theory to Practice: Applying Graph Theory Network Analysis, 2024.
- [25] Stanford. Lecture notes. Lecture notes, 2025.
- [26] Wikipedia. Closeness centrality. Closeness centrality, 2024.
- [27] Wikipedia. Label propagation algorithm. Label propagation algorithm, 2024.
- [28] Wikipedia. Betweenness centrality. Betweenness centrality, 2025.
- [29] Wikipedia. Centrality. Centrality, 2025.
- [30] Wikipedia. Dendrogram. Dendrogram, 2025.
- [31] Wikipedia. Hierarchical clustering. Hierarchical clustering, 2025.
- [32] Wikipedia. Jaccard index. Jaccard index, 2025.
- [33] Wikipedia. Jensen–shannon divergence. Jensen–Shannon divergence, 2025.
- [34] Wikipedia. K-means clustering. K-means clustering, 2025.
- [35] Wikipedia. Louvain method. Louvain method, 2025.
- [36] Wikipedia. Modularity (networks). Modularity (networks), 2025.
- [37] Wikipedia. Multinomial logistic regression. Multinomial logistic regression, 2025.
- [38] Wikipedia. Network theory. Network theory, 2025. Consulta en línia.
- [39] Wikipedia. Random forest. Random Forest, 2025.
- [40] Wikipedia. Singular value decomposition. Singular value decomposition, 2025.
- [41] Wikipedia. Spectral clustering. Spectral clustering, 2025.
- [42] Wikipedia. Total variation distance of probability measures. Total variation distance of probability measures , 2025.